January/February 2015

InTech

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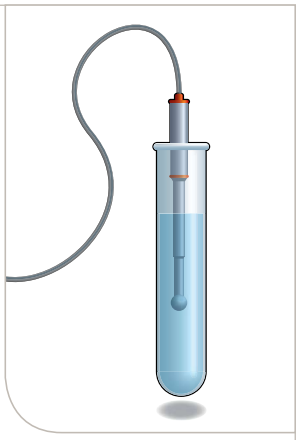
WEB EXCLUSIVE

Measuring pH of ultrapure water



Measuring the pH of ultrapure water is challenging; water in this pure state is one of the most aggressive solvents known, dissolving virtually everything to which it is exposed. It requires specialized sensors, meticulous installation, and proper maintenance.

Read more at:
www.isa.org/intech/201502web

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ISSN 0192-303X

***InTech* is published bimonthly by the International Society of Automation (ISA). Vol. 62, Issue 1.**

Editorial and advertising offices are at 67 T.W. Alexander Drive, P.O. Box 12277, Research Triangle Park, NC 27709; phone 919-549-8411; fax 919-549-8288; email info@isa.org. *InTech* and the ISA logo are registered trademarks of ISA. *InTech* is indexed in Engineering Index Service and Applied Science & Technology Index and is microfilmed by NA Publishing, Inc., 4750 Venture Drive, Suite 400, P.O. Box 998, Ann Arbor, MI 48106.

Subscriptions: For members in the U.S., $9.52 annually is the nondeductible portion from dues. Other subscribers: $155 in North America; $215 outside North America. Multi-year rates available on request. Single copy and back issues: $20 + shipping.

Opinions expressed or implied are those of persons or organizations contributing the information and are not to be construed as those of ISA Services Inc. or ISA.

Postmaster: Send Form 3579 to *InTech*, 67 T.W. Alexander Drive, P.O. Box 12277, Research Triangle Park, NC 27709. Periodicals postage paid at Durham and at additional mailing office.

Printed in the U.S.A.

Publications mail agreement: No. 40012611. Return undeliverable Canadian addresses to P.O. Box 503, RPO West Beaver Creek, Richmond Hill, Ontario, L48 4RG

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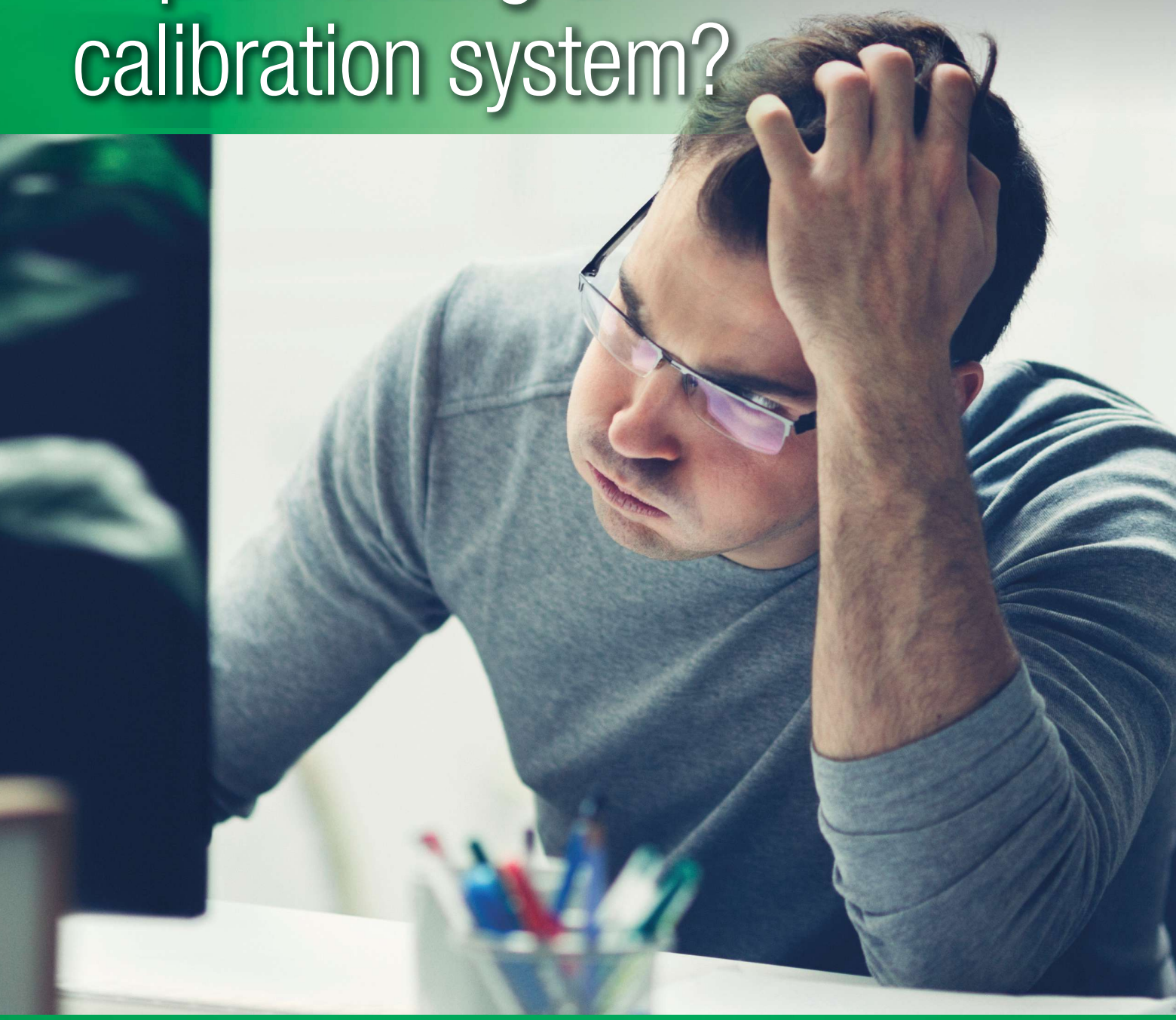
InTech provides the most thought-provoking and authoritative coverage of automation technologies, applications, and strategies to enhance automation professionals' on-the-job success. Published by the industry's leading organization, ISA, *InTech* addresses the most critical issues facing the rapidly changing automation industry.

Implementing a calibration system?



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Act or react? Make it a decision

By Bill Lydon, *InTech*, Chief Editor

Someone told me years ago it is not important how you react to problems, issues, and challenges, but how you act. This idea is very powerful. Generally our instinct is to react quickly to problems, issues, and challenges, but this is not always the best approach. Obviously if the building is on fire you need to get out, but fortunately most things we face in life are not that urgent.

After using this idea for years, I finally did a little research and discovered it was derived from the Greek philosopher Epictetus, who lived about 1,900 years ago. Epictetus taught it is not what happens to you, but how you react to it that matters. He promoted philosophy as a way of life and not just as a theoretical discipline. To Epictetus, all external events are determined by fate, and are thus beyond our control; we should accept whatever happens calmly and dispassionately. This can be difficult in many situations that rouse our emotions. He also believed individuals are responsible for their own actions, which they can examine and control through rigorous self-discipline. This is sound life advice, and I think it applies to engineering and automation, particularly when we run into problems. A variant on this philosophy is what my AC and DC fundamentals professor in college would instruct us when posing a problem, "where does the reasoning begin?" When a problem or challenge arises, this is another good question to consider.

Epictetus taught it is not what happens to you, but how you react to it that matters.

Many times when faced with an automation or control problem or challenge the tendency is to react with a quick-fix solution. Assuming the problem is not threatening safety or property, there can be great value in avoiding the temptation of implementing a quick fix. Step back and think about what is happening before developing a solution. This can lead to higher quality solutions that are process improvements, eliminating future problems.

It is also valuable to ask others for their thoughts. Engineers and technicians in particular prefer to solve the puzzle themselves as a matter of pride and have difficulty asking others for opinions or help. My perspective comes from first doing engineering and later managing engineers. In many cases you end up reinventing the wheel. It is more important to get things done using all the resources available. Resources include coworkers, suppliers, and fellow ISA members. Online communities such as the ISA group on LinkedIn allow you to post questions to the over 30,000 members in the automation field.

It is not important how you *react* to problems, issues, and challenges, but how you positively *act*. Actions speak louder than words, but reasoned actions based on sound information add lasting value. ■

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Alarm management

I just read the alarm management update ["Auditing alarm management systems," November/December 2014 *InTech* web exclusive, www.isa.org/intech201412web01]. As a longtime member of the ISA18.2 committee and also co-chair of two of its seven TR [technical report] committees, I wanted you to know that I thought the update article was very well done. In particular, I appreciated the content regarding evolving the ISA-18.2 standard to an IEC [International Electrotechnical Commission] standard and also the good coverage of the several ISA18.2 committees developing TRs.

Joseph Alford, P.E., CAP

The right generation?

I challenge Dr. Martin to consider that his reflection of generational differences is not linked to the times ["The right generation at the right time," September/October 2014 *InTech*]. Rather it is the same generational passing that people have seen since the beginning of recorded history. Labeling a group based upon its contemporary hip dynamic, such as "millennial" or "baby boomer," to characterize the group a certain way simply does not pass muster. Surely it's popular to assign characteristics to certain groups as if the cited generation is some historically unique and never-to-be-repeated phenomenon.

What I've seen—having passed through my first 64 years—is entirely age related. Everyone is claiming erudite status by attempting to label groups and fit normal generational change into some sort of tech-related cubbyhole.

The young are always perceived as incapable, unmotivated, lacking in values, and just not up to the task. The myth that youth means more creativity, affinity for new technology, or ability to cope simply lacks statistical demographic proof. There are youthful sloughs, techno illiterates, and Luddites, as there are tech-savvy, innovative, and solutions-oriented mature workers.

The characteristics attributed to youth of today were pasted onto my generation in the '60s and '70s. The explanation is we all aged, gained experience, suffered failures, and enjoyed victories. Because of that, aging altered our habits, practices, and beliefs. And yes, there were, are, and will be outliers. The characteristics cited in the article are more about age than when people were born. I can take any paragraph and remember how the label was applied to the few generations I've lived with. Read the first sentence of each and fill in the blanks.

So the burden and advantage of youth is to be explorative, ignorant, seeking, nonconforming, and quick. For older people, it's to be wise, experienced, knowledgeable, and understanding. In between, we get a mix and match of the two. As technology advances, some pick it up and others eschew change—but that's not a generational imperative, nor is it an age-group mandate.

What's not discussed is *leadership*. Leadership is a necessary ingredient for organizational, industry, and national success. Those in charge must mentor youth, model good behavior to all, and master new developments. That has been true from the dawn of time. It's true now and will—I'm confident—be true tomorrow.

Do we capitalize on those generational advantages? You bet we do. Because good leadership—after modeling correct behavior for others to emulate—requires knowing the masses and tailoring our guidance to what will resonate with them.

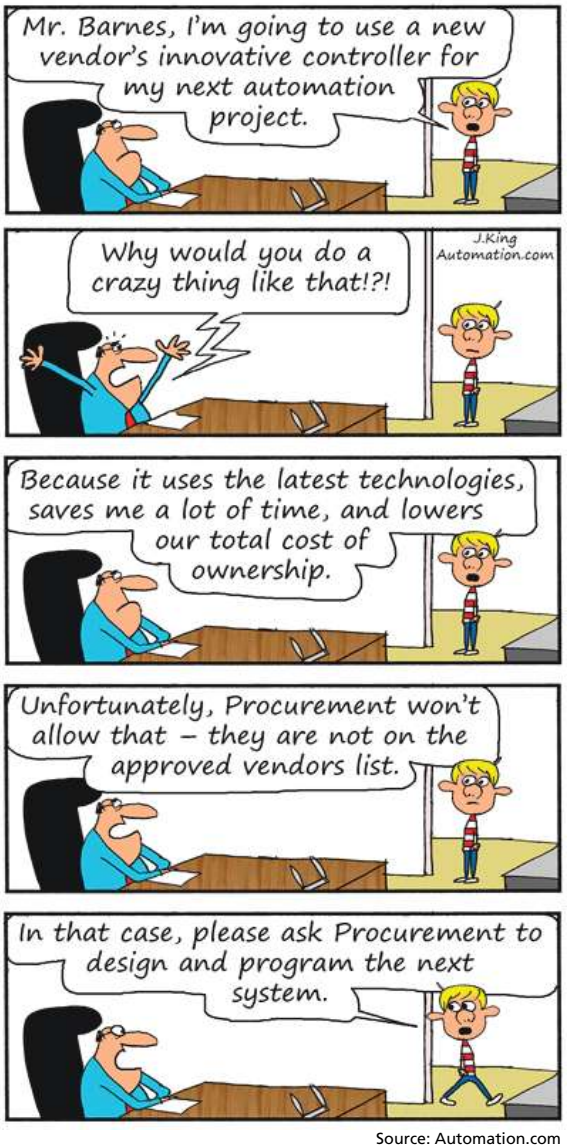
Charles Drobny

Educate for technology

I am a boomer and work with a millennial at a power generation facility as I&C [instrumentation and control] techs; it is a pleasure to work with him. He was hired as an apprentice with minimal education or experience in instrumentation and control, but with an education in Ethernet networking. He is now a journeyman and is doing well.

My concern is our educational system—which does not seem to promote education in technology with an awareness of the technology jobs that are available. I have worked as an engineer for more than 20 years and now make more money with better benefits as a technician. In my area, companies are cutting back on engineers as they retire and depending on outside contractors to do the engineering; when it is incorrect, the technicians get to fix it. Change is inevitable; how many blacksmiths were without work when the automobile entered our society? I believe the key is to keep the money in our society. We cannot do this unless we are competitive and innovative.

Jeff Warner



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New law confronts rising risks of cyberattack

The U.S. Congress approved and President Obama signed The Cybersecurity Enhancement Act of 2014. The bipartisan act strengthens and protects the nation's economic and national security through public-private partnerships to improve cybersecurity and to create a greater reliance on cybersecurity standards, research and development, workforce development and education, and public awareness and preparedness.

Passage of the bill follows years of efforts to pass federal cybersecurity legislation. An earlier cybersecurity bill, The Cybersecurity Act of 2012, was defeated in the Senate. Its demise prompted President Obama to instruct the National Institute of Standards and Technology (NIST) to develop the U.S. Cybersecurity Framework, which was introduced in February 2014.

At the federal government's request, representatives of both the Automation Federation and ISA served as expert consultants to NIST as it coordinated the development of the U.S. Cybersecurity Framework. In fact, long before the President called for a federal initiative on cybersecurity, Automation Federation and ISA leaders were consulting with White House National Security staff, U.S. federal agency officials, and members of Congress on the critical need to establish national cybersecurity standards, guidelines, and compliance testing.

Industrial automation control systems security standards developed by ISA (ISA99/IEC 62443) are integral components of the federal government's plans to combat cyberattack, because they are designed to prevent and offset potentially devastating damage to industrial plant systems and networks—commonly used in transportation grids, power plants, water treatment facilities, and other vital industrial settings.

The Cybersecurity Enhancement Act of 2014:

- Authorizes NIST to facilitate and support the development of voluntary, industry-led standards and best practices for critical infrastructure—drawing on many of the key recommendations in the U.S. Cybersecurity Framework
- Strengthens cybersecurity research and development by building on existing research and development programs and ensuring better coordination across the federal government
- Ensures the next generation of experts are trained and prepared for the future
- Increases the public's awareness of risks and cybersecurity
- Advances cybersecurity technical standards ■

Reshoring efforts slowing

Global strategy and management consulting firm A.T. Kearney released its *2014 Reshoring Index,* the first in a series of studies looking objectively at the rate and pace of the return of manufacturing operations to the U.S. In this inaugural index, manufactured goods flows are tracked over 10 years to show the change in ratio between U.S. manufacturing imports and gross output during the period. The index is actually expected to show a year-over-year decline, lower by 20 basis points from 2013, as offshoring to foreign manufacturing markets outpaces reshoring.

"While the so-called reshoring trend has helped improve the mood of U.S. manufacturing since the recession, the reality is that the import value of manufactured goods into the U.S. from 14 low-cost Asian countries has grown at an average of 8 percent per year in the last five years," said Pramod Gupta, A.T. Kearney principal and study coauthor. "The *2014 Reshoring Index* is not only an indicator of U.S. manufacturing capital flows, but also how the U.S. stacks up in terms of attractiveness as a source of manufactured products versus countries like China, Bangladesh, and Cambodia." Highlights include:

- The top three reshoring industries (as measured by the number of cases in A.T. Kearney's database) are electrical equipment, appliances, and component manufacturing, with 15 percent of the cases; transportation equipment manufacturing, with 15 percent; and apparel manufacturing, with 12 percent.
- Improvement in delivery time led the reasons executives gave in favor of reshoring, with quality improvement a close second, and brand/image third.
- Although there has been an overall lift in U.S. manufacturing for five straight years since 2009, imports of offshored manufactured goods into the U.S. have increased at a faster rate than any return of manufacturing operations. ■

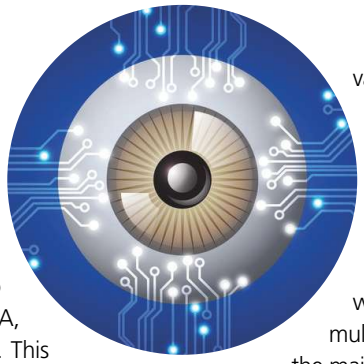
AT&T collaborates with Rockwell Automation

Rockwell Automation and AT&T are collaborating to improve remote asset utilization and connected machine management. The companies will help Rockwell Automation customers securely collect, manage, and act on data from industrial equipment located in plants and remote sites around the globe by bringing the AT&T Global SIM and M2X Data Service platform together with Rockwell Automation cloud-enabled service offerings.

The companies are also developing reference architecture and validated designs for cellular Internet of Things deployments within industrial sites. These tools will guide customers in the design and use of cellular connected assets in ways that complement the existing in-plant network infrastructure. ■

Automation by the Numbers

$1.65 billion



Total sales of machine vision components and systems increased 12 percent to **$1.65 billion** in the first nine months of 2014, according to new statistics issued by AIA, the industry's trade group. This is the best nine month start by the North American machine vision market since AIA began tracking quarterly statistics in 2009.

Total machine vision sales include sales of machine vision components and systems. Total machine vision components are up 18 percent to $230 million through October, while total machine vision systems are up 12 percent to $1.42 billion. The third quarter of 2014 was especially strong for machine vision systems, which posted growth of 19 percent. Machine vision components also increased in the third quarter by 2 percent over 2013. Every component and system category increased in the third quarter of 2014 except for lighting and imaging boards. Smart cameras (44 percent), optics (16 percent), software (14 percent), and application-specific machine vision systems (14 percent) were the standout individual categories in terms of year-over-year growth for the third quarter of 2014.

"Despite the tempered expectations of AIA membership after the strong first half of the year, the market continued to grow in the third quarter," said Alex Shikany, AIA's director of market analysis. "Our most recent survey shows that AIA members expect machine vision components to increase and systems to remain flat in the next six months." ■

3,000,000

HMS Industrial Networks has shipped its **3 millionth** Anybus module to Bosch Rexroth. Bosch Rexroth, which has been working with HMS for more than 10 years, uses Anybus modules mainly to connect welding and tightening controllers to the various control systems used in automation technology. Anybus embedded products are interchangeable communication modules for different fieldbus and industrial Ethernet networks. The modules have multi-network connectivity for the major network standards.

The Anybus CompactCom communication modules consume little power and are compact (5.2 cm x 5.2 cm). The standardized hardware and software interfaces allow the modules to be easily interchanged with each other. In other words, you can connect to any network by simply replacing the module. ■

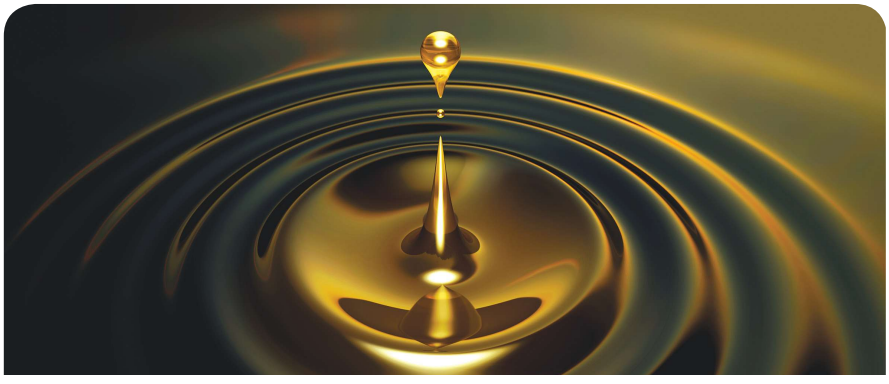
$2.0 billion

Flow Research says the market for Coriolis flowmeters totaled $1.3 billion in 2013 and is projected to grow to almost **$2.0 billion** by 2018. Growth in the energy markets, especially in oil and gas, is creating greater demand for the accuracy and reliability of Coriolis flowmeters. Coriolis flowmeters remain the most accurate flowmeter made, and both accuracy and reliability are critically important for measuring the flow of crude oil and petroleum liquids. Although Coriolis flowmeters are used for both upstream and downstream petroleum applications, they are especially suited to downstream applications.

Another important force driving the market is the development of large line size Coriolis flowmeters. For many years, nearly all Coriolis flowmeters were used in pipes with diameters of 6 inches or less. In the past five years, four major suppliers have developed Coriolis flowmeters for use with pipes from 8 to 16 inches in diameter. Even though these flowmeters can be expensive, they are becoming increasingly popular. Most of these large line size Coriolis flowmeters are designed for custody transfer applications.

Coriolis flowmeters have also benefited from industry approvals that previously worked mainly in favor of differential pressure and turbine flowmeters. The American Gas Association approved a report on the use of natural gas for custody transfer applications in 2003. This report helps explain the growing use of Coriolis flowmeters for natural gas. The American Petroleum Institute (API) has also issued a draft standard for the use of Coriolis flowmeters to measure hydrocarbon fluids. This document was added to the API Library in July 2012. The API also approved a draft standard called *Measurement of Crude Oil by Coriolis Meters.*

Even though Coriolis flowmeters are being used more widely to measure both natural gas and industrial gases, liquids still account for more than 75 percent of Coriolis flow applications. And even though measurement of the flow of petroleum liquids is growing at a faster rate than measurement of nonpetroleum liquids, measurement of nonpetroleum liquids still represents a larger segment of the Coriolis fluid measurement market. ■



Plant historians

Historians provide data to gain insights and improve operation

By Bill Lydon

Plant historians became popular in process plants in the 1980s. More recently they are being used broadly throughout industry to improve operations, identify problems, and find opportunities for enhancement. An early forerunner of historians were electromechanical pen-and-paper chart recorders. They produced a graph of one or more measured values over a period of time for analysis and a permanent record of critical information.

Plant historians acquire real-time data from automation and other systems to store time-stamped data at high speed to maintain a chronology of information. This industrial process information is then available to any user for reference and analysis including production status, performance monitoring, quality assurance, troubleshooting, tracking, and genealogy. A major advantage of historians is the ability to research and correlate any data easily to identify trends and relationships.

Initially historians were very expensive, but they have now gained broader use. Lower-cost computers and storage make it possible to record large amounts of plant data from operators, sensors, and processes at a reasonable cost. Compared to keeping paper records, it is much simpler to manage data, analyze it, and archive it electronically.

Big data

Before the term "big data" was being used as part of the Internet of Things, plant historians were handling large volumes of time-synchronized data. Consider time sampling and storing a single temperature sensor value every half second (500 milliseconds)—it generates more than 63 million samples per year. Tracking 1,000 temperature sensors in a plant would generate more than 63 billion stored samples per year. For sequential samples where values do not change, data compression techniques are often used to conserve storage. It is also common to filter measurements before sending data to a historian to reduce the significant amount of "noise" in the data. In an automation system, typical items tracked and stored include temperature, flow rate, pressure, level, machine cycles, run time, and overall equipment effectiveness. The recognition that big data can be valuable has put greater emphasis on historians to capture more data in plants than in the past. Fortunately, technology enables the collection and handling of big data at lower costs.

FAST FORWARD

- Plant historians store time-stamped data at high speed to maintain a chronology of information.
- Historian information is used for analysis to find patterns and make improvements.
- Plant historians were handling large volumes of data before the term "big data" become popular.

Legacy electromechanical chart recorder

Leveraging data

Information increasingly is being put to use by a wide variety of people, including plant management, engineers, operators, accountants, business analysts, scientists, quality control workers, and information technology (IT) specialists. Data alone does not deliver benefits; it is the people who use the information to solve problems and make improvements who deliver benefits. The ISA-95 international standard for the integration of enterprise and control systems data has been widely adopted to organize and communicate historic information. There are several use cases for data captured by historians:

Legal and compliance verifications

In both process and discrete industries, companies need to maintain a genealogy record of production and quality tests for legal and compliance reasons. Historians have detailed data that can be used in defense against litigation and to determine what products must be recalled, if necessary. Requirements in the food and pharmaceutical industries are long standing. Recent quality incidents and recalls in the automotive industry illustrate the need to capture and keep production data. Pharmaceutical application of historians must comply with 21 Code of Federal Regulations (CFR) Part 11, also known as electronic records/electronic signatures. Given the importance of the data, the FDA has strict regulations regarding access, security, and ability to edit such records, as documented in this CFR.

Track and trace serialization

Track and trace (TnT) serialization initiatives to maintain the history of products in the pharmaceutical and food industries are creating a greater need to capture and retain historic production records. The pharmaceutical industry is gearing up to TnT as government agencies and companies take measures to reduce drug counterfeiting and product diversion and increase patient safety. There are a number of motivations including contamination recalls, counterfeit drugs (Viagra is the most counterfeited drug), and drug thefts. The street value of drugs can easily range from $15 to $50 per tablet or in the case of codeine, $200 to $300 per pint. Thefts of pharmaceuticals in transit have ranged from $2 million to $80 million per incident. Countries worldwide are planning to require track and trace, increasing the need for plant history information.

Root-cause analysis

When production issues occur, historic data is fundamental to identifying sources of problems using root-cause analysis. A root cause is a cause that prevents the final undesirable event from recurring when it is removed from the problem fault sequence. For example, common issues in process applications that can be identified include operator error, inadequate cleaning times, poor cleaning solution strengths, device malfunctions, and wrong process temperatures.

Troubleshooting

Many times, having chronological data is fundamental to pinpointing automation, control, and equipment problems. Chronological data gives troubleshooters deep insight into the behavior of processes at any point in time.

Optimization

Chronological historic data provides in-depth system performance analysis that companies use when developing methods to optimize processes. This information can be the inputs to simulation software to understand the production process and simulate optimization methods.

At the annual Pharmaceutical Automation Roundtable, a major pharmaceutical company described how it uses historian data for new insights to improve the control methods of biological processes that are complex and difficult to predict. Using historic information and off-the-shelf analytic software, it learned about the interaction between process and equipment data that had not been apparent in the past. This also helps to identify a range of other problems, including maintenance issues.

Power and energy monitoring

Energy has been a blank check in industrial plants. However, by using a historian, along with submetering and power monitoring, plants can allocate energy costs to production steps and products to achieve closer cost accounting, find problems, and identify areas for improvement.

Predictive analytics

Advanced intelligence capabilities are being applied using data trends and patterns to predict failures and events. These predictive analytics are only possible with historian data.

Investment justification

Justifying the investments to replace and upgrade equipment can be more accurate using actual historic plant data, opposed to estimates. Historic data provides integrity to investment proposals made to management.

I recently asked an automation manager from a major pharmaceutical company about the value of historians, and he commented they are invaluable for providing the data store needed for true process understanding based on analysis, especially for batch processes. He also noted that many managers think of historians as simply a data store to cover a firm for regulatory purposes and do not clearly see the untapped value of information that can be harvested from historians for process improvement, process optimization, or aiding with root-cause analysis.

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Identifying data

There are different philosophies about how much data to capture and store in a historian. One theory is to capture every possible data point, since you never know what you will need in the future. This seems a bit extreme. The goal is to determine in each process or production area what would be needed for analyzing long-term performance and what would be needed for identifying immediate and short-term problems. It is probably more productive to discuss ideas about what data to capture from various departments based on their needs. Creating lists of questions for which people want factual answers can help stimulate ideas for data to capture.

The application of data is an ongoing activity to explore and understand. Initially users may not know enough about the data to put it to productive use, and over time they may recognize the need to capture more data points. This is the idea of "peeling the onion." As you learn more, there are typically more questions to research.

Wireless sensors are making it more cost effective to monitor and capture data in historians without installing wiring, leading to new operational insights.

Embedded historians

There is a new breed of embedded historians in controllers and standalone field devices that collect historic data remotely. These are rugged field hardware devices with solid-state memory that are part of controllers, plug into backplanes, or communicate through industrial networks to capture data, time tag it, and store it. Data can be captured immediately at the source and time tagged, making it more accurate. It is forwarded to the corporate or cloud database for long-term storage. This store-and-forward method allows data aggregation, so the central historian database transaction does not need to occur synchronously to the sampling rate. Onboard software rules engines may be incorporated into these devices, which can be configured to perform analysis for optimizing processes and production.

Having the absolute time relationship between data is critical to proper operations or analysis in some applications.

Having the absolute time relationship between data is critical to proper operations or analysis in some applications. For these applications, a number of controllers have options for more precise time stamping. Controllers that support the IEEE 1588 standard (Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems) can communicate with a precision time reference. Another method is for the controller to use a global positioning system radio receiver for a highly accurate time reference.

Using the open OPC UA industrial Web services standard, the information can be sent to historians, business intelligence databases, and enterprise systems, and queried over the Internet or in-house networks.

Database interfaces

Historian databases are high-performance, optimized designs to capture and time tag data at high speed. Sending this data to a computer industry-standard database structure, such as SQL, allows users to take advantage of a wide range of commercial reporting and analysis software. In addition, pushing information from historians to cloud servers is a way to store a large amount of data and lower cost without having to own, maintain, and manage more servers.

Cloud analytics

An interesting development is the availability of analytic software tools and analysis platforms as cloud applications at a very reasonable cost. Another advantage of this approach is it leverages high-performance computers to accomplish analysis more efficiently.

Business intelligence and historians

Historians and business intelligence (BI) systems have developed independently, but there is a growing recognition that they need to be linked. Historians are unique since they can chronologically capture high-speed, real-time data. In the business world, the time of transactions and events are typically measured in minutes, hours, or days rather than milliseconds. Historians and BI systems share the goal of capturing data to provide historical, current, and predictive views of operations for reporting, online analytical processing, analytics, data mining, process mining, complex event processing, business performance management, benchmarking, predictive analytics, and prescriptive analytics. The goal is to have a resource for information that people can easily access and use.

The new challenge is integrating the silos of dissimilar data to transform it into useful information. Having a plan to link systems and provide a framework that allows teams to combine isolated "islands of information" (including building management, plant utilities, process control, production, and business data) allows users to view and analyze disperse data in a variety of ways, leveraging their standard automation infrastructure.

Developing a master plan with an integrated way for users to leverage data from multiple systems (i.e., historians, business intelligence, MES, LIMS) will achieve the goal. The solution should connect to other databases and systems to access data elsewhere without replicating the master data. Standard interfaces to other data sources provide the capability to bring in data from other sources and monitoring tools. In addition, it is an advantage to allow access to this information via the Web using handheld devices as well as laptops and desktop PCs.

One user told me his vision is that users need to be "two clicks away" when using a tool to browse to the data. He wants to make the data easily available to the right people using Web browsers with proper access control through the

company intranet. This will provide links to knowledge management tools and contextualized data from various applications (ERP, LIMS, QTS, historians, etc.). More accessible data will be used to improve operations. Examples include data for analytics colleagues to build models and relevant data for continuous improvement champions. Ad hoc reporting is an important functionality for this system to enable people to focus on specific issues and investigations.

The solution should use commercial off-the-shelf technology to benefit from widely accepted industry standards. This will ensure timely migration to the latest technology, while avoiding large expenses and the cost of custom coding and maintenance.

Cooperation

Knowledge is power if you harness it and make it available to make better decisions and improve automation. Doing this takes cooperation between people in the silos of your organization. History has taught that increasing access to information enables people to gain insights and make improvements. Knowledge is dynamic, so the systems need to be responsive to the users and extendable over time.

Having sound data is only the starting point; it takes knowledgeable and skilled people to use it. The technology implemented should have the goal of lowering users' time to gain insights. Outside of a few early adopters, technology by itself has negative value to the majority of people. Most people are afraid of change, and it is important to educate users.

The tremendous growth of IT investments is accelerating and creating a range of off-the-shelf software for tapping data sources, analyzing big data, and closing the loop to optimize business operations and processes, including manufacturing. The overall goal of closing the entire loop for business operations through manufacturing enabled by the Internet of Things may well be the next force driving the integration of IT and automation. The plant historian role is to be the single location to capture and store large amounts of real-time data. Big data concepts are knitting together silos of data more holistically to improve business operations. In the manufacturing and process industries, the plant historian is an important data source, along with distributed data located in automation controllers and devices. ■

ABOUT THE AUTHOR

Bill Lydon is chief editor for *InTech*. Lydon has been active in manufacturing automation for more than 25 years. He started his career as a designer of computer-based machine tool controls; in other positions, he applied programmable logic controllers and process control technology. In addition to experience at various large companies, he cofounded and was president of a venture-capital-funded industrial automation software company. Lydon believes the success factors in manufacturing are changing, making it imperative to apply automation as a strategic tool to compete.

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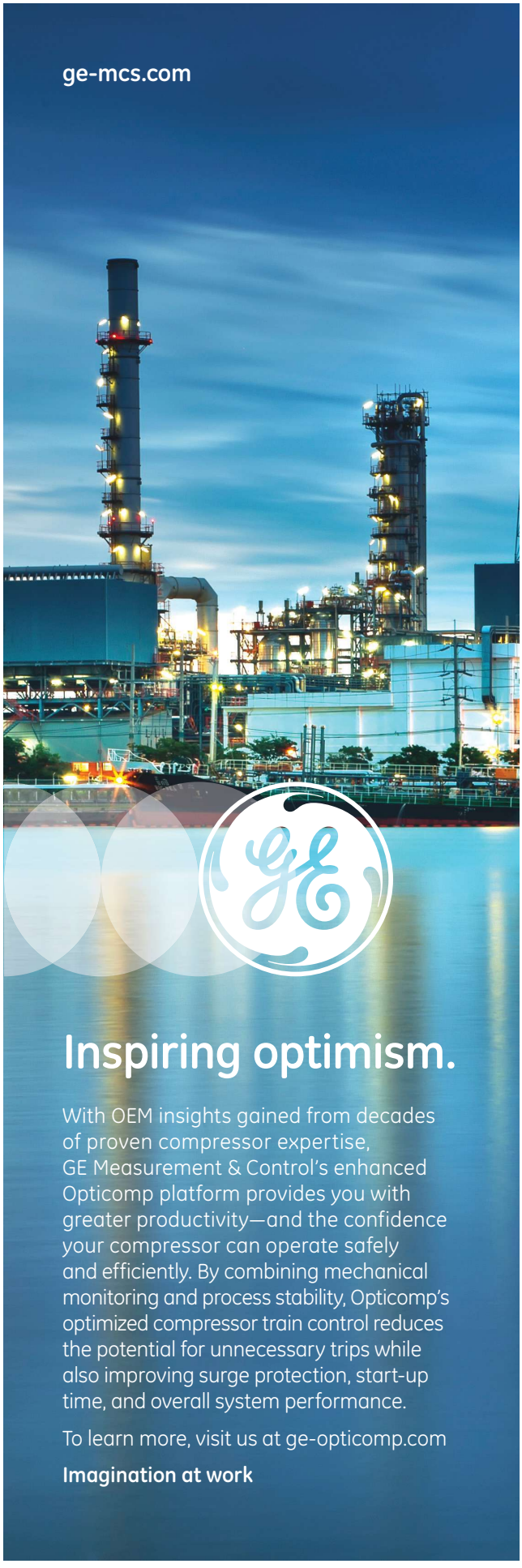

Basic control functions may keep a process stable, but much more is needed to optimize performance and achieve specific company goals

Advancing from process control to process automation

By Paul Darnbrough, P.E., CAP

The way we have started automobiles through the years is a simple analogy to the progress, or lack thereof, of process automation technology. Car owners have been carrying keys for the better part of a century. At first it took two operations to start a vehicle: the key turned on a switch, and the driver stepped on the starter. Later, starting was performed as a single action with the key. Later still, cars added central locking functions, still performed with a mechanical key.

As technology advanced, functions performed with a key became more complex, until the key itself was made largely obsolete. Basic control functions, such as locking the doors

and starting or stopping the engine, became more sophisticated. Now the car is able to sense the owner (or at least the owner's key fob) approaching, and it unlocks the door as he or she grasps the handle. Once inside, the system provides a secure means of push-button starting the vehicle, and it may even go so far as adjusting the mirrors, seat position, and entertainment settings to meet a specific driver's preset desires. One could say the basic functions of opening, starting, and adjusting a car have been advanced and elevated into the realm of advanced automation.

Process industries have developed along similar lines with control systems. To answer the question of what process control is exactly, we have to go back to the earliest introductions of control mechanisms, where first-generation electro-pneumatic-mechanical loop controllers replaced people doing tasks such as manually adjusting valves in response to some local indicator like a pressure gauge.

Although a device was used to automate a human function in an effort to control a variable, there was no sense of what the process was doing overall. A basic controller could keep an individual loop on an even keel, more or less, so long as there was not too much disruption. Complex processes might employ dozens or even hundreds of such controllers, each with its performance displayed on a panel board, but keeping an eye on the big picture was still a human process.

Moving to electronic control

When distributed control system (DCS) platforms were introduced in the 1970s, they simplified the mechanics of the panel board, but did not do much to improve its capabilities. Big-picture analysis was still largely a human responsibility. Sure, getting beyond the technical constraints of pneumatic field devices with their troublesome compressed air tubing made it easier to install more instruments and actuators, but the basic control concepts did not really change. Any movement to advanced process control (APC) and other forms of control optimization were still in their infancy. Process automation capable of supporting APC had to encompass many technologies and techniques. It was characterized by incorporating many more input data points into algorithms and orchestrating more complex sequences.

Older systems did have powerful capabilities available to those willing to explore them. Some sophisticated users were operating with fundamental APC concepts even back in the pneumatic era, but those successes required a high degree of internal engineering capability. There were few, if any, tools available commercially to support such efforts. The same applied to early DCS platforms. Few companies ever overtaxed the brute computing power of the processors running a DCS, but creating the kind of programming necessary to drive APC in such an environment was no small task.

Figure 1. Even the most modern process plants typically do not take full advantage of the capabilities of their control systems.

The hard work of optimization

The transition to process automation and APC was empowered by being able to create an all-encompassing platform capable of coordinating more than single loops or small cascade groups. One major advantage of newer platforms is the ability to optimize a process to suit the owner's specific economic goals based on any number of desired outcomes. The process automation system can operate the plant to minimize energy consumption, maximize output, and deliver specific product quality attributes. Companies using these systems effectively swear by their capabilities.

Implementing such systems is challenging, and having an automation solutions provider working with an internal engineering department can make the task much easier. During the initial design phase of a control system

FAST FORWARD

- Even the most modern process plants typically do not take full advantage of automation system capabilities.
- Newer control systems and smart instruments can be harnessed to automatically optimize plant processes and overall operations.
- Many plants can benefit from outside assistance when moving from basic process control to full process automation.

upgrade or a new installation, it is far too easy to focus just on process fundamentals, and never get beyond considering desired steady-state conditions. Automation system upgrades and new installations can therefore miss opportunities to engage with process and automation technology experts capable of uncovering better ways of doing things.

Bringing in fresh ideas

An automation solutions provider can bring new eyes and ideas to advance a project beyond what designers conceived initially. While the individuals within a given plant may understand their plant processes intimately, such a group may not have the time to go beyond current capabilities. In some instances, these individuals may also lack broader knowledge of automation systems, particularly as applied to processes in other plants.

One of the major advantages of bringing in outside talent is tapping the collective knowledge of a larger group of engineers who have worked on many projects in many environments. Each new experience adds to the knowledge base, and it can be transferred as part of a planning process. Even a question as simple as, "Why is this control action performed in this manner?" can prompt discussion and cause companies to consider new and better ways of performing routine functions.

Many capabilities of modern process automation systems are still underutilized in most process plants, even among companies most people would consider sophisticated (figure 1). Far fewer companies use APC as effectively as they could, even though basic APC technologies have been around for decades.

Even fewer have developed systems for implementing procedure automation to deal with startups, grade changes, shutdowns, and other disruptions—even though such situations are the primary causes for process upsets and safety incidents due to the high degree of human intervention involved and the infrequency with which they occur. The ISA-106 standard covering procedure automation may be relatively new, but the concepts embodied in the standard have been around for many years.

As a purely practical matter, human capabilities and the skills of experienced operators are indispensable to operating a plant well, but too many plants are overly dependent on unwritten tribal knowledge. A review of reports analyzing process safety incidents will turn up many situations where an inadequately trained operator had to take manual control of a process during a startup or other changeover, and ended up making the wrong decisions. Companies lose huge amounts of money in such situations.

Properly developed process automation systems are always on the lookout for trouble, and are ready to respond and alert operators when a problem is anticipated or detected. More advanced control sequences stand ready to be executed via procedure automation, even if they are only used once per year. Comprehensive process automation systems cannot only handle plant operations automatically, but can also supplement operator knowledge and activities by supplying the right amount of information at the right time to the right people.

Capturing operator knowledge

Automating actions through procedure automation is an excellent way to capture tribal knowledge and the understanding of a plant's best people before they retire or move on (figure 2). The need for operator training remains, but procedure automation reduces dependence on human memories and an individual's ability to make the right decisions in a crisis. Control systems, even relatively old ones, can perform such functions when programmed properly, but outside assistance may be required to incorporate this functionality.

As we deal with the "great shift change" driven by worker demographics, the ability to automate the entire range of process control functions through procedure automation will become even more important. Experienced long-time operators often have a wealth of unwritten knowledge regarding plant operations waiting to be captured and automated. The technologies exist; it is a matter of taking up the challenge and doing it—and automation solutions providers can help.

More devices, smarter devices

Another area where a higher level of sophistication in process automation is critical relates to the increasing numbers of smart devices and systems in process plants, both wired and wire-

Figure 2. Procedure automation and other techniques can capture tribal knowledge from a plant's best operators.

Table 1. Smart valve information transmitted to control system

- Precise position
- Time spent in a given position
- Opening and closing force
- Stiction and binding
- Process noise
- Number of actuations

less (figure 3). The quantity of modern field devices offering extensive reporting and diagnostic capabilities has grown by orders of magnitude, as has the information each can deliver. These devices are easily networked via a variety of protocols, which provide a huge pipe for delivering mass quantities of data.

No longer does each device provide a single 4–20 mA signal corresponding to the process variables, as now there is status information about a transmitter's health or a valve's condition (table 1). In fact, the flood of information can be too much of a good thing if not handled correctly.

However, a well-configured process automation system is capable of harvesting what may seem to be an overload of data, then digesting it to make it useful. From a process standpoint, extended process data can be boiled down to established key performance indicators, which in turn feed back to optimize operations.

Careful consolidation of this data into control room or mobile dashboards gives operations personnel at-a-glance visibility into the system status.

Modern process automation components also have functionality beyond what is needed to directly control or automate a process, as they now often supply valuable data to maintenance management systems, historians, mobile devices, and so on.

Figure 3. Smart instruments like this wireless guided wave radar level transmitter supply much valuable information beyond the process variable measurement.
Source: Emerson Process Management

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Table 2. Benefits of moving from basic control to advanced automation

- Facilitates process optimization
- Applicable to normal steady-state operation
- Can be applied to disruptive operations like startups and shutdowns
- Enhances worker safety with quick responses to unusual situations
- Efficiently supplements human intervention
- Captures knowledge from a retiring workforce
- Integrates well with advanced smart device information
- Natural fit with maintenance management systems, historians, and mobile reporting to identify issues

More than the sum of the parts

Making all these elements work together to create a symbiosis of technologies and work processes is a daunting task. Choosing the best approaches from the dozens or hundreds of possibilities in a given situation can seem overwhelming, and may cause some companies to remain in the past for fear of investing too heavily in wrong technologies or applying the right ones ineffectively. An automation solutions provider can help users sort through seemingly endless options and make appropriate choices.

Once those choices are made, all the individual elements have to be networked together to support optimized interaction. This is where the participation of an automation solutions provider is critical, as control systems and components are selected and implemented to connect disparate parts into a seamless whole. These activities depend on the accumulated know-how of engineers and technicians who have worked with a variety of major platforms, countless subsystems, and numerous plant processes.

Companies that have implemented major projects thoughtfully with careful planning and help from a capable automation solutions provider typically realize better performance, reduced costs, improved safety, and other benefits (table 2). Having automation systems capable of controlling plant processes without constant human intervention creates a much safer environment, and allows a company to thrive even in the face of changing and challenging conditions. ■

ABOUT THE AUTHOR

Paul Darnbrough, P.E., CAP, is principal engineer for MAVERICK Technologies. He has more than 20 years of experience in engineering, documentation, and construction of automated industrial and process control systems. Darnbrough has worked with clients ranging in size from small single-owner operations up to Fortune 500 companies and government agencies on processes including plastics, food, chemicals, material handling, discrete manufacturing, water treatment, and pharmaceuticals.

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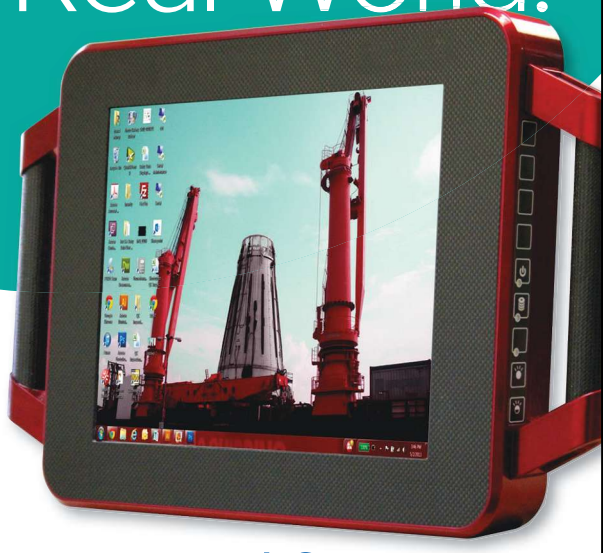

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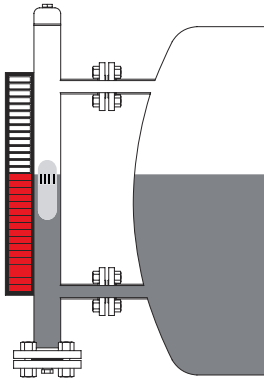

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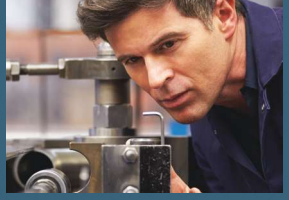

MAJOR UPGRADE = MAJOR MENTORING OPPORTUNITY

Don't ask: Who can do this the quickest? **Ask:** Who will benefit the most from doing this?

Major control system upgrades and migrations are rare. Even if an operator or engineer spends their entire career at the same facility, their exposure to these types of projects might be limited to only one or two occurrences. And once they retire, their knowledge of the process and control system leaves with them.

As we actively battle the growing skills gap in manufacturing, we have to look for opportunities to transfer personal and tribal knowledge that is otherwise difficult to capture. Control system upgrades offer some of the most teachable moments out there.

70%

OF ESSENTIAL JOB SKILLS ARE LEARNED

ON THE JOB

DCS Migration Has It All

From individual field devices to the enterprise level, the main process control system touches everything. Consequently, virtually all aspects of manufacturing operations and associated automation systems are addressed with a typical DCS migration — and so are most of the standard project management stages. There's no better way to expose new engineers to increasingly complex areas of project management — particularly financial planning, IT functions and security concerns — than by fully immersing them in the reality of these projects. Though expediency is often prized over education, there's a wealth of information to be gained here if we make time for it. The key is to focus on training opportunities rather than the finish line.

Connect the Dots

When new engineers and operators are starting out, it can be a challenge to visualize how the individual elements of a process — valves, reactors, instruments, pumps and so on — relate to each other. But to create an automation system that operates consistently and profitably, this is exactly the type of knowledge they need.

Involvement in a migration offers a unique, real-world opportunity to gain a deeper understanding of the process. Device-by-device analysis of field instrumentation lends insight into how the process was initially designed. Reviewing the hazard picture can reveal weaknesses that need to be addressed. And studying communication networks and operator interfaces helps build a solid foundation for design best practices. Every step is a chance to learn something new.

INVOLVED FROM START TO FINISH

Most projects follow a fairly predictable series of steps. Involving new engineers every step of the way helps establish a valuable knowledge base and provide context for future projects.

1. **Brainstorming**
   Imagine and assess the new system's potential
2. **Creating Budgetary Estimates**
   Understand the value and cost of each item
3. **Analyzing Legacy Drawings & Code**
   Gain a historical perspective of the process
4. **Shooting Control Loops**
   Learn how to analyze and optimize operations

Training a New Generation

Right now, the demographic landscape for automation engineers is split fairly evenly between baby boomers and millennials. The former are quickly approaching retirement, and the latter are just entering the workforce. But this is not the only way in which they differ. Millennials don't think or learn the same way baby boomers do. They were raised alongside technology and, as such, have expectations of its capabilities and its potential. Hands-on training is one of the most effective ways to involve the next generation of automation leaders early on so they can help shape the industry's future.

How to Make the Most of It

Of course, it's rare for a company to have the in-house engineering capacity to facilitate its own migration. Most turn to a control system supplier and / or an automation system integrator for assistance.

MAVERICK Technologies, like many systems integrators, helps companies with their migration projects, and also drives workforce development efforts in conjunction with these projects. Its DCSNext solution takes a holistic approach to DCS migration that integrates plenty of opportunities for collaboration and training.

MAVERICK isn't the only company to follow this approach, but not all do, so it's important to ensure that your service provider include training as part of your DCS upgrade. This is the type of investment that will pay dividends for decades, so make the most of it.

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Remotely managed industrial automation cybersecurity

Continuous monitoring and operational assets protection

By John Crawford

FAST FORWARD

- Does your organization have the operational technologies, processes, and security expertise necessary to adequately prevent, detect, and respond to the latest malware targeting industrial automation control systems?
- Ongoing cybersecurity protection is a moving target requiring continuing investment.
- A remotely managed industrial automation cybersecurity service is an alternative that leverages technology and experts to protect operations from cyberthreats.

Epic Turla, Regin, and BlackEnergy are the names of just a few in a series of recent malware discoveries that target industrial automation control systems (IACSs). The increasing volume and sophistication of these types of threats has driven the topic of operational technology security all the way up to the corporate boardroom. Unfortunately, the attack capabilities being developed by these adversaries is outpacing the defensive capabilities target organizations are adopting.

Does your organization have the operational technologies, processes, and security expertise necessary to adequately prevent, detect, and respond to the latest malware targeting industrial automation control systems? Ongoing cybersecurity protection is a moving target requiring continuing investment. An option is engaging a managed security service (MSS) to improve your operational security posture. This article will help you understand the key benefits and challenges of an MSS, as well as some criteria to consider in selecting an MSS provider for your operational security.

Benefits

There are four primary benefits of a managed security service:

- Predictable cost
- 24/7/365 coverage
- Access to qualified domain expertise and technology
- Disentanglement from noncore competencies

Compared to starting up and staffing a private security operations center (SOC), outsourcing is significantly more cost efficient.

The challenge here is access to competence and qualified industrial cybersecurity (ICS) resources. Even if your organization already has an enterprise SOC, or if cost is not a primary factor for your organization, it can be difficult to hire qualified IACS security professionals because there are so few available. On the other hand, training staff with the necessary skill set can be a time-consuming and onerous process for those new to the automation domain.

For mission-critical IACSs, 24/7/365 security monitoring and management is advisable. A managed security service provider can supply trained and qualified staff to support round-the-clock monitoring and response. In addition, it will likely also have access to technologies that enable a higher quality of service than you could provide using in-house capabilities.

Although going with a managed security service can have tremendous benefits related to cost, coverage, and quality, do not undervalue gaining the freedom for the business to focus on core competencies and operational excellence. By matching your operational technology security requirements to the appropriate service-level agreements, you can provide the right level of security while maintaining peace of mind regarding operational performance expectations.

Not a silver bullet

However, subscribing to a remotely managed security service is not a silver bullet. There are some organizations that rely on the network isolation of their IACS to provide an additional layer of security. If there is no remote connectivity to the IACS, for obvious reasons, it cannot be remotely monitored or managed. However, be wary if there is also no local security monitoring or management being performed. By itself, an air-gapped IACS network is insufficient to provide an acceptable level of security. In reality, a network-isolated IACS environment may be exposed to greater risk than that of a securely connected and remotely monitored IACS.

Systems with remote connectivity also may not be a good fit for continuous remote monitoring if the network uplink is already highly utilized, performance constrained, or expensive to operate. Therefore, it is important that the managed service vendor has the necessary consulting and professional services in its portfolio to enable secure operations without requiring always-on remote monitoring and management.

A common trap to look out for is the false assumption that remote access to the control system always equates to increased risk. Secure remote access technologies are mature and proven in the industrial space. Many IACSs already have remote access functionality to enable support, engineering, or maintenance. The decision to use a managed security service provider should be based on what skills, know-how, processes, and resources become available by allowing the remote connection, compared to relying solely on in-house capabilities.

Selecting managed security service

Selecting the right managed security service provider for your operational environment can become an onerous task, so here are some things to consider that will help with the evaluation process.

Does the provider understand operational technology and have the capability to secure the environment without degrading operational performance, usability, safety, maintainability, or serviceability? One of the most important criteria to consider when evaluating a managed security service provider is its maturity in adapting enterprise information technology (IT) security technologies to the operational technology space. Especially in legacy systems, it is very easy to cause the control system to malfunction or to introduce incompatibilities due to ignorance or carelessness. Even with seemingly primitive security controls, like antivirus, a configuration mistake can suddenly and unexpectedly bring a workstation's performance to a screeching halt.

Because of this risk, the best security partner for you and your service provider is an organization that understands and has working experience with industrial automation systems. Industrial automation systems are different than business IT systems, and most vendors publish a list of security products that have been tested and verified to be compatible with their automation systems. The vendor may even provide specific configurations or limitations that need to be considered in an ICS. Any service provider you consider should have a solid understanding of your automation system vendor's product and patch compatibility, security guidelines, and how to effectively use any security functionality integrated into the solution. A key partner is a domain operator who can identify the challenges associated with the unique problems in ICS. ***Does the provider have global capabilities?***

If your organization operates globally, it is an advantage to use a managed security service provider that can give uniform services to your organization. The alternative is dealing with multiple providers and the increased cost of managing fragmented service from multiple

The decision to use a managed security service provider should be based on what skills, know-how, processes, and resources become available by allowing the remote connection, compared to relying solely on in-house capabilities.

organizations. In addition, you should ensure that the provider maintains local capabilities where your major operational assets reside.

Do the provider's service level agreements address your operational performance needs? Establishing the performance metrics that you will use to gauge your quality of service should be communicated and agreed upon as early as possible when establishing a relationship with a managed security service provider. This will be a guide to determine the fitness of the provider to your organization's specific needs. It is also important to understand what recompense you can expect if your service provider fails to deliver the agreed-upon service levels. These metrics usually include service availability, mean time to remediation, mean time to deploy antivirus updates, etc. The important thing is to choose metrics that encourage the behavior you want from your service provider.

Does the provider's portfolio align with your service needs? Most managed security service providers offer the following services: security and risk consulting, firewall management, intrusion detection system (IDS) management, log monitoring, security information and event management, and vulnerability management. However, there are also some operations-specific pain points you may like addressed. Typically, these are things like configuration and policy deployment, device and signature updates, proactive vulnerability alerting and patch notification, vulnerability remediation, regulatory or compliance auditing, or vendor-specific security services like custom signatures or device hardening.

Is the provider reputable? Industrial automation control systems are very long lived. You should partner with a managed security service provider that will be around for at least as long. Ensure that the provider has a good reputation and is financially stable.

Is the provider's deployment model compatible with your business needs? There are certain service design elements that may be important to clarify, based on the business's requirements. For example, the pricing model may have to accommodate flexibility in focusing more on capital expenditure or operational expenditure, depending on how the business prefers to handle expenditures. Other deployment issues relate to where equipment and data will physically reside, who has management responsibility for what, and how the transition from unmanaged to managed operational technology will be handled.

What distinguishes providers? Cybersecurity protection is a moving target, and it is advantageous to have a service provider with a portfolio of people, technology, and ongoing development focused on industrial automation cybersecurity. For example, service providers producing custom IDS signatures based on ongoing in-house vulnerability research can provide greater protection. Advanced forensics capabilities are built upon know-how of industrial automation systems including human-machine interface software, controllers, protocols, and other devices. Most often, this distinguishing technology is in the form of in-house developed and refined methods and algorithms used within the security operations center that allow for more accurate analysis, a more sophisticated way of achieving a result, or an efficiency improvement through automation.

A major value of using a managed cybersecurity service is the people in the operations center. The employees keep up to date on the growing number of cyberthreats and the tools to deal with them, which makes them proficient and efficient. ■

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John Crawford is a control system product and solution security expert with more than 15 years of engineering experience. He is an innovator in the domain of industrial automation security, with multiple patents pending. Crawford is sought after for his diversity of expertise in the areas of security design, security architecture, secure software development life cycle, product security testing, security governance, risk management, and compliance.

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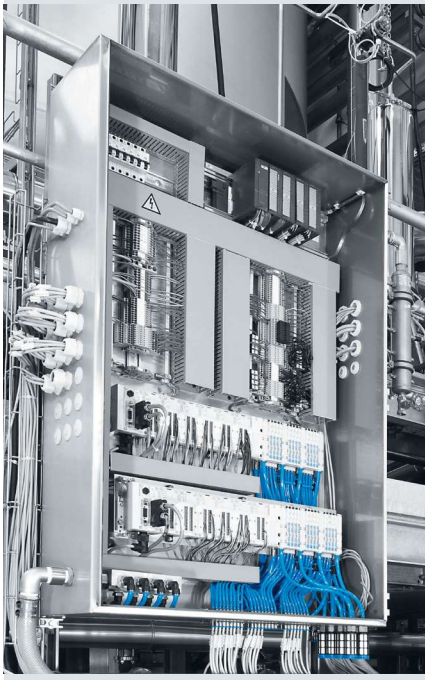

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Fieldbus technology has been available for more than 20 years. Initially, FOUNDATION Fieldbus and other digital fieldbus technologies were developed in the 1990s as a replacement for the 4–20 mA standard. Today, virtually every manufacturer of flowmeters, pressure transmitters, and similar instrumentation offers FOUNDATION Fieldbus and other popular fieldbus interfaces, and every major automation system vendor supports one or more fieldbus standards.

More recently, ISA-100 and WirelessHART were also developed. The International Electrotechnical Commission (IEC) approved WirelessHART in March 2010 as IEC 62591, and ISA-100 was approved in September 2014 as IEC 62734. For the purposes of this article, *wireless* refers only to wireless sensor systems, and not to other wireless technologies such as 802.11 Wi-Fi.

Today, automation professionals in a process plant have a choice to make for new installations: wired or wireless? For plants with existing wired and wireless infrastructure, the choice hinges on a straight comparison of the two technologies and the application of the solution that makes sense. For plants without an existing wireless infrastructure, the cost of installing one must be considered (figure 1).

This article shows the advantages of each technology, recognizing that almost every plant will end up with a mix of wired and wireless. To simplify comparisons, FOUNDATION Fieldbus and WirelessHART will be used as leading examples of fieldbus and wireless technologies.

Wired versus wireless trade-offs

How to choose for new installations

By Moazzam Shamsi

Each has significant competitors, but comparison of competing technologies in each area is outside the scope of this discussion.

Fieldbus details

Compared to traditional 4–20 mA wiring, fieldbus technologies save wiring costs, simplify expansion, and are easier to make redundant because they allow multiple instruments to use a single cable called a *trunk* or *segment*. A trunk or segment begins at an interface device at the automation system. On a FOUNDATION Fieldbus system, the interface is called an H1 card.

The DC power needed for instruments on a FOUNDATION Fieldbus segment is provided by a power supply rated up to 500 mA, enough to theoretically power more than 32 instruments. In practice, however, 12 to 16 instruments are typically installed on a segment. Some instruments require more than 20 mA; available power diminishes over long cable lengths; and engineers like to allow capacity for adding instruments. Typically, up to 12 devices can be installed on a fieldbus segment up to 120 m long. If the process unit has more than 12 instruments, a second or third segment can be installed.

If a problem occurs in any instrument on the segment—such as a short circuit—it can disable the entire segment. Therefore, many plants install a segment protector or device coupler, allowing multiple instruments to connect at one location. The device coupler is installed in an enclosure near the process unit. Connections to the individual instruments are called *spurs*.

Figure 5. Steam trap monitoring with Rosemount 708 wireless acoustic/temperature transmitters allows accurate detection of issues before they escalate.

FAST FORWARD

- Automation professionals working in process plants can choose between wired fieldbus or wireless for many new projects.
- Some visionaries predict all-wireless plants are coming, but this is not possible yet.
- Wireless will gain in popularity because it can monitor anything, anywhere.

Figure 1. Many process plants have both wired fieldbus and wireless infrastructure in place, allowing the use of either for new installations and projects.

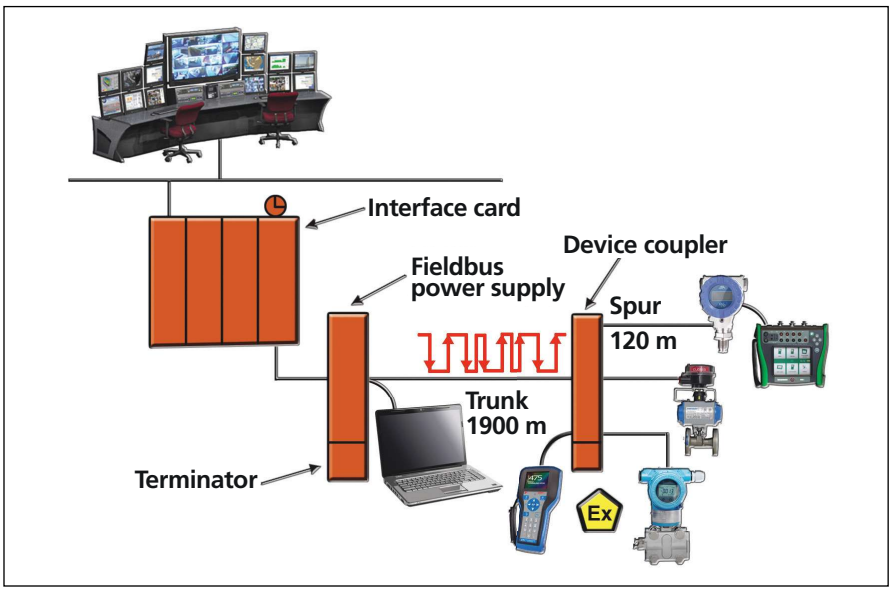

Figure 2. FOUNDATION Fieldbus diagram. Multiple FOUNDATION Fieldbus instruments can be connected to each instrument through the device coupler and then networked back to the host control system.

A typical segment includes an interface card, a fieldbus power supply, a device coupler, and individual spur cables from the coupler to the instruments (figure 2). With FOUNDATION Fieldbus, the H1 card communicates to the plant's distributed control system (DCS) via a high-speed Ethernet (HSE) connection. Other fieldbus technologies use similar architectures.

For critical process units, a redundant segment can be installed, using duplicate segment cables and power supplies. In hazardous areas, intrinsic safety barriers provide protection.

Where do you use Fieldbus?

The ideal application for wired fieldbus is a process unit containing many flow, pressure, temperature, level, multivariable, and other instruments, all within a reasonable distance of each other. The more instruments in a relatively small area, particularly complex multivariable units, the more fieldbus makes sense (figure 3).

By using device couplers and marshalling cabinets strategically located around the unit, wiring from instruments to device couplers can be minimized. Most fieldbus instrument suppliers offer automated design tools, making it easy to design a segment, calculate maximum distances, and determine the wire types. When instruments on a segment are far apart, repeaters allow segment distances up to 300 m.

Instrumenting such an application is fairly easy, because multiple vendors make components with various fieldbus interfaces. If a plant wants to modernize its legacy control system and install fieldbus-based instrumentation, HART can use existing 4–20 mA wiring from older instruments to carry digital information to the device couplers. The device couplers can be installed in the old marshalling cabinet, saving a considerable amount in wiring and labor costs. Although HART does not have performance levels comparable to newer fieldbus technologies, it is the least costly wired digital option, and often sufficient from an operational standpoint.

Valve and pump controls are also available with fieldbus, so it is possible to set up local control loops within the fieldbus array operating independently from the DCS. This is accomplished with function blocks allowing, for example, local proportional, integral, derivative (PID) control of a digital valve controller based on signals from a nearby level transmitter. If the DCS or the HSE go down, the control loop will continue to operate.

Shanghai Wujing Chemical, an acetic acid plant in Shanghai, China, upgraded its controls and instrumentation to increase its capacity from 300,000 tons/yr to 530,000 tons/yr. The new system included an Emerson DeltaV DCS and FOUNDATION Fieldbus instrumentation. Shanghai Wujing

Figure 3. These two Rosemount 3051S multivariable transmitters are each sensing temperature and pressure, and transmitting this and other information back to the host control system via FOUNDATION Fieldbus.

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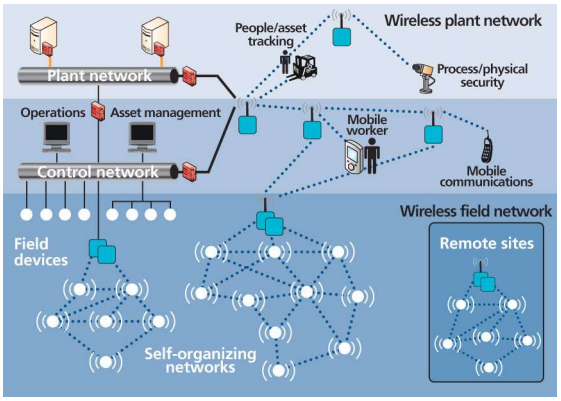

Figure 4. Modern wireless networks can be used to communicate with many types of devices spread over widely dispersed areas.

used local control for 114 PID loops. This saved 74 percent of the DCS controller process time. FOUNDATION Fieldbus diagnostics and communications are now used during calibration of control valves, saving 80 percent of the time previously needed for maintenance and operations.

Typically, wired fieldbus devices and segments have more power available than wireless transmitters. This makes wired fieldbus suitable when working with loop-powered devices such as two-wire level transmitters with continuous wave modulation, eight-channel process temperature transmitters, tank gauging multispot temperature transmitters, intelligent on/off valves, and field indicators.

Wired fieldbus is also suitable for real-time process control. The response speed of a wired system—from the spur to the segment to the H1 to HSE to the DCS and back again to a control component, such as a control valve—can be significantly faster than a wireless system.

Because of the cost of the hardware (H1 interface, power supply, cable, etc.), wired fieldbus is not suitable for a few devices located far away from other instrumentation. Such applications are better handled by wireless transmitters, along with other scenarios such as adding instruments to existing plants without installed extra capacity in the wired infrastructure.

Wireless details

WirelessHART is a self-organizing mesh technology in which field devices form robust wireless networks to dynamically mitigate obstacles in the process environment (figure 4). Other wireless technologies employ similar strategies, with varying degrees of effectiveness.

Wireless technologies do not require communication wiring and related infrastructure. Some wireless devices require power wiring, but the vast majority of deployments use battery power and thus operate completely without wires. Wireless networks communicate data back to host systems securely and reliably, and can be applied to both control and monitoring applications.

A WirelessHART installation requires, first of all, a wireless instrument transmitting data according to the IEC 62591 WirelessHART standard. If a transmitter does not have WirelessHART, a wireless adapter can be installed on most existing two- or four-wire devices.

With WirelessHART, each wireless device transmits to a gateway managing a specific wireless field network. Typically, the gateway is assigned to a process unit. Each gateway manages its own wireless field network and can have an assigned HART tag like any HART device. Each wireless field network in a plant has a unique network ID to prevent devices from attempting to join the wrong network.

WirelessHART devices can communicate through each other to send messages to the gateway, forming a self-organizing "mesh." The mesh extends the range of a device beyond that of its own radio. For example, a wireless device may be several hundred feet away from the gateway with obstacles between, but power-efficient "hops" through neighboring devices closer to the gateway ensure a reliable, extended range. The gateway connects to the DCS via a wired or wireless high-speed link, typically Ethernet.

Where do you use wireless?

The chief advantage of wireless systems is that they can be installed virtually anywhere in an efficient, timely, and cost-effective manner. Battery-powered transmitters require no wired infrastructure or local power supply, so they can be far away from a process unit's wired fieldbus and power wiring. They can also be installed in locations where supplying power and cabling would be too expensive or hazardous. This flexibility means that there are benefits to using wireless in both greenfield capital projects and existing brownfield facilities.

Greenfield capital projects typically want to make 10 to 20 percent of traditionally wired signals wireless. Engineering, procurement, and construction contractors and plant owners see strategically incorporating wireless as beneficial in terms of reducing physical fixed infrastructure. They also use wireless technology to manage schedule risk and cost escalation/containment and to reduce space requirements and weight.

Wireless can reduce schedule and cost impacts because there is always the potential for scope creep and additional I/O as projects progress. The later in the project a change comes, the greater the risk of jeopardizing the project schedule and budget. Wireless can usually accommodate these changes better than fieldbus.

Quite often, additional measurements are eliminated from the design of a new or retrofit project because those signals are deemed too costly to implement during the capital expenditure phase of a project. If those measurements are needed later, adding wired instrumentation can be much more costly than wire-

Table 1.
Advantages of Fieldbus

- Widespread support among instrument suppliers
- Supported by almost every automation system supplier
- Familiar to maintenance personnel
- Excellent for control
- Decades of operating experience
- Best for multivariable instruments and multipoint devices
- Digital data shows raw measured signals
- Extensive diagnostics
- Less wiring than 4–20 mA

Table 2.
Advantages of wireless

- Significant reduction in installation complexity
- Inexpensive
- Reduces space, weight, and power requirements
- Easy to expand to accommodate changes and additions
- Excellent for monitoring applications
- Can be used for control in some circumstances
- Best for hard-to-reach locations
- No need to modify existing wired infrastructure
- Can reduce the need for fixed asset inspections, reference to IEC60079 Part 17
- Allows measurements to be made where fieldbus is not practical or feasible

less solutions. On these types of brownfield projects, wireless solutions are a good way of supporting operational excellence programs to improve plant productivity and reliability and compliance to emerging environmental and safety legislation.

A plantwide wireless network can be installed with minimal disruption to fixed infrastructure such as wiring, conduits, and cable trays. Adding wiring to these existing assets can be costly, and interfering with aging assets can cause unforeseen issues such as disrupting signal transmission. For example, a major chemical company in Europe implemented a plantwide wireless infrastructure to enable operational expenditure programs. The first application was an energy project to monitor steam traps. The company installed wireless acoustic transmitters to listen for acoustic signatures and monitor temperature—both vital signs related to steam trap operation (figure 5). The steam trap monitors took only two days to install, and they generated a complete return on investment (ROI) in six weeks through detection and remediation of previously unknown losses.

In another application, a global contractor reported that adopting wireless solutions cut commissioning time by 50 percent. It also streamlined the design and engineering process, saving 10 hours per measurement point. Using wireless helped them maintain the project schedule and improved their ability to manage change on the project.

A third leading wireless application is fixed asset inspections per IEC 60079 Part 17, which requires plants to continuously monitor electrical equipment in hazardous areas. This directive obliges maintenance personnel to manually check instrument cable glands, cables, terminal field junction boxes, safety barriers, trunking, ducting, pipes, and conduit for leaks, corrosion, tightness, and similar problems. Wireless transmitters do not require wired infrastructure, thus eliminating this costly maintenance task.

The low-power nature of WirelessHART devices allows them to operate for several years without replacing the battery. Configurable update rates conserve battery life by choosing the most appropriate rate for a particular application, typically within a range of 1 second to 1 hour. Most applications today are "monitoring" and therefore only need infrequent updates. WirelessHART transmitters can in certain circumstances be used in real-time control applications, but this requires higher update rates, in some cases necessitating wired power.

Many projects have a very attractive ROI when using wireless that they do not have with wired fieldbus due to its added installation and maintenance costs. In these situations, wireless can quickly deliver an investment payback, with ongoing benefits.

Using fieldbus and wireless together

Some visionaries predict all-wireless plants are coming, but this is not possible yet. Instead, modern automation platforms and connectivity standards are making it easier to use fieldbus and wireless together. Those designing new automation projects should consider using both as complementary technologies in a tactical way to efficiently manage projects and to enable operational excellence programs.

Fieldbus and wireless can be used together to manage project risks, such as I/O scope creep, and engineered to meet power, space, and weight design constraints. Ideally, a basis for design should be established early in the project so engineers clearly understand which technology to use in each situation.

Wired fieldbus will continue to be used, because it can perform real-time process control with a DCS and with control-in-field function blocks and provide signals and power to control devices. Wireless control is possible with WirelessHART using PIDPlus algorithms, but it will take time for industry to gain the confidence and experience to apply this technology for real-time control.

Wireless will become more popular because it can monitor anything, anywhere. Also, it is more economical than wired fieldbus and can be easily installed in both new and old plants. Industry-leading process plants will install both wired and wireless infrastructure and use whichever is most efficient and effective in each situation. ■

ABOUT THE AUTHOR

Moazzam Shamsi, M.Sc., B.Eng., C.Eng., MinstMC, has been an automation professional for 25 years, and his career spans a broad range of industries and roles from technical leader to project management. He presently works for Emerson Process Management where he globally directs Emerson's wireless consulting and execution solutions on large capital projects. Shamsi specializes in working with clients and contractors to implement technology solutions for capital and operational efficiency projects.

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RESOURCES

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"Projects commissioning with Fieldbus Foundation"
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"Industrial wireless sensor networks"
www.isa.org/standards-publications/isa-publications/intech-magazine/2012/october/web-exclusive-industrial-wireless-sensor-networks

Doing RFID right

By Steve Loyal

Avoiding 12 common missteps

Adopting radio frequency identification (RFID) asset tracking for the first time involves more than just slapping tags on traceable materials. It requires research, expert support, and a well-thought-out plan.

RFID can be a huge boost to a company's productivity and reliability. RFID can generate valuable real-time data that helps a company avoid product and equipment shortages, ensure reliable, on-time delivery or service, track parts inventory, provide maintenance history in the field, and much, much more. But like any technology, those embracing it for the first time should know why they are doing it and how it can improve their business, not just what it costs. When implemented properly, with realistic goals and the right technology, RFID pays for itself quickly, again and again. Doing it right means recognizing and avoiding some critical mistakes—common ones—that can leave a new adopter regretting the initial RFID investment and leery of spending more.

These are what I call the dirty dozen—12 mistakes that can make an RFID investment go bad—and how to avoid them.

1. RFID is not for everyone

RFID has come a long way. It is the right step forward for many businesses. But not all. Some may see it as must-have technology, without having a firm grip on how it would improve their businesses. Perhaps a competitor is using it. Or they like being early adopters. That is well and good, but not enough to justify an initial RFID investment. Step one in making a good RFID investment is knowing why it is right for you. How would you use the intelligence the data provides? Are you ready to change your operations, even if that costs money, too? Might another technology do just as well for now, with less cost and complexity? For example, do you do any electronic asset tracking now? Do you use bar coding, and if so, how well has it been working for you? Is it now falling short? These are all good questions to ask yourself to avoid making a poor RFID investment.

2. Pain point not clearly defined

It is fine to have general goals of improvement. Every manager wants to optimize his or her company's operational efficiency and asset management. However, when considering a RFID system, some first-time adopters go in without an understanding of what they are trying to accomplish and end up buying too much or too little RFID. What pain points do you want to address? Are you losing, or misrouting, too many assets? Is it hard to find inventory in your plant or warehouse? Have you incurred costs, or faced unscheduled downtime, because the incorrect tool was inserted into a machine, damaging one or the other? Whatever specific pain you are experiencing should be analyzed, quantified, and given a dollar cost value. You should be able to estimate the theoretical savings and later benchmark actual savings achieved by adopting RFID.

FAST FORWARD

- Companies should avoid these common mistakes for a superior return on investment.
- RFID software investment should never be an afterthought.
- Most companies are not up to installing their own RFID systems.

3. ROI estimate is inaccurate or incomplete

Now that you have an accurate picture of how much your pain points are costing you, estimating the return on investment (ROI) and time frame for achieving it should be straightforward. Should be, but it is not always. Hardware is sexy. Software is not. Focusing on hardware alone can lead to seriously underestimating how long it will take to recover an RFID investment. A complete RFID system is a capital investment. The majority of the investment is the software and integration portions, not the hardware. An accurate ROI calculation will take all costs into consideration, including training and potential operational changes to make good use of the data. Once all of this has been considered, it may become a capital investment decision the chief financial officer makes. In most cases, it will not be a major investment, and the ROI will be fully captured in fewer than 18 months. Do not let a single outside party do the ROI for you. Do it yourself, or even better, have multiple parties give an estimate. That way, you can be sure all angles have been considered and a consensus ROI figure emerges.

4. Not understanding what the technology can do

Some new adopters are confused about RFID's capabilities. They often have a preconceived idea that they can slap a smart label on their inventory, tooling, or shipping containers, and they will be traceable automatically at all times. That is not true. A successful RFID system requires many properly matched and integrated hardware and software components. Do some research about the technology. Ask the people selling RFID lots of questions. The time to understand what the technology can and cannot do is before you sign the purchase order.

A complete RFID package includes tags, antenna, a reader or readers, and a middleware package and/or software suite.

5. Not involving the IT department

Also, do not wait until the last minute to bring your information technology (IT) department into the RFID discussions. Their involvement is critically important. RFID, after all, is all about data. The data has to get through their system and come out in a useful format. This is IT's turf, and the importance of their buy-in to the process, system, and output cannot be understated. Respect the maxim regarding RFID projects: It is not serious until IT is involved.

6. Not considering the correct frequency RFID

Now, let's get a little more technical. The choice of frequencies refers to the size of the radio waves passing between RFID components. The wrong choice may add cost or introduce inefficiency. Passive ultrahigh frequency (UHF) technology is the most versatile choice now, capable of covering a broader read range—and hence supporting more applications—than ever. Where historically high frequency (HF) technology was used for short distance read ranges, the same results now can be successfully obtained using UHF technology. Today's smaller UHF antennas and passive UHF tags have excellent short read ranges, down to inches, or long read ranges even beyond 50 feet. This can make passive UHF an attractive alternative to either HF or active UHF.

7. Active or passive RFID? You can go wrong

An internal battery powers active RFID tags, and the radio waves from the interrogating device power passive RFID tags. With active RFID systems, there are a couple of potential pitfalls to keep in mind. First is the expected life of the batteries in the transponders (tags). In your proposed application, will it be practical to change the batteries when they need replacing? More often than not, it is simpler to change the tags. This is an expense, both for new tags and the labor time to do the job. Also, your Wi-Fi system may not be able to cope with active tags. Such an RFID system may overpower an existing Wi-Fi network, leaving both systems battling for bandwidth. When it comes to active RFID, consider whether you would be better off upgrading your Wi-Fi, too.

8. Not all tags are robust enough for their environment

Finding the right tag is critically important to the success of your RFID project. Yet, some adopters fail to make sure their tags are tough enough for all the operating and environmental conditions to which they will be subjected. The wrong tag will not only jeopardize the success of the project, but could provide a false impression about the overall effectiveness of the technology. Consider the operating environment, and find tags tough enough for it. Will they be exposed to harsh temperatures, a lot of water, or high humidity? Will the tags need to operate on metal or in a metal-heavy environment? Or are they destined for nonconductive surfaces? What read range is needed? Is extended memory needed? Among the many options for tags today you will find solutions to any application-specific requirements!

9. In choosing hardware, there is such a thing as too little or too much

Choosing the wrong antenna and reader is another common misstep. You should consider the size and strength of the antenna to make sure it will align with the read range needed. Select an antenna that allows you to read all the tags desired and not more or less. Standard patch antennas are the norm, but there are new technologies on the market that can be ideal for tougher applications such as server racks, doorways, and smart shelves. Readers come in many different configurations as well and can be selected to fit your application. Options include the number of antenna ports, power levels, and the antenna range. Larger, extended-range readers have many features to cover most applications, with all the bells and whistles. For some new RFID adopters, buying such readers may seem like future-proofing. However, the technology is getting better all the time, and businesses might be perfectly well served starting with a smaller, short-range reader. As they say, there is no need to bring a bazooka when a fly swatter will do the job.

HARTING's Ha-VIS LocField traveling wave UHF antenna shows how new technology is expanding the reading zones and hence the flexibility of RFID in specific applications, like those involving doors and pass-throughs or small shelves.

10. Making the wrong software decision

There are very good options available for RFID software. Choose carefully, because the wrong choice can be limiting, or at the other extreme, overload users. Configurable middleware packages offer an easy-to-use bridge from the reader to your core IT system. Many expert RFID integrators supply excellent middleware and software packages. Also, there are options for application-specific software for handheld readers. This can be important for users in the field. A system will be even more efficient when that user does not have to think about how to manage the data; the application software does it for them. Such a system will be easier to master and foolproof.

11. Not using an expert integrator

Maybe this should be the first on our must-not-do list, but assume you have taken the trouble to do all of the internal assessments of needs and learned the RFID basics. That makes you an informed first adopter. However, putting together RFID hardware and software to create a properly functioning system is a systematic process involving many sequential steps. Often companies try to install RFID themselves only to have it fall short of expectations because of poor implementation. Then the technology gets blamed. The burden of implementing RFID is going to fall on your IT department. So ask yourself this, and be honest, "Are we up to doing the job ourselves?" In almost all cases, the answer is no. The solution is to outsource RFID integration. An expert integrator can analyze the application, optimize the hardware, provide the appropriate software, implement and maintain the system, and even train staff. Such an integrator will bring you efficiency and results that exceed your expectations.

12. Not doing a proof of concept

Last but not least, do not go too far too fast. Take a phased approach to your RFID implementation. Start with a small manageable portion of the project. Perform thorough testing at each stage of the implementation. Let the system run for a while so you can identify any needed adjustments. And be flexible. You may need to change the reader or antenna that seemed like the best choice before starting. Before relying on your RFID network, satisfy yourself that everything is functioning as it should by doing a proof of concept (POC). The POC will help ensure everything fits together in practice like it did on paper.

As you can see, there is a lot of depth and scope to selecting and implementing a good RFID system. Keeping all these factors in mind upfront should result in a smoother process and improved ROI. ■

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RESOURCES

Wireless Networks for Industrial Automation, Fourth Edition
www.isa.org/store/products/product-detail/?productId=115943#sthash.ZKWff1BD.dpuf

"The ABCs of RFID"
www.isa.org/link/abc_rfid

"RFID unites the supply chain"
www.isa.org/link/rfidunites

"RFID yes, UPC no and probably not"
www.isa.org/link/rfid_yes

"Tracking wireless: Experts answer users' questions on tracking wireless assets in industrial settings"
www.isa.org/standards-and-publications/isa-publications/intech-magazine/2008/april/special-section-wireless-tracking-wireless

"On the radio"
www.isa.org/standards-publications/isa-publications/intech-magazine/2006/june/factory-automation-on-the-radio

Justifying automation in the era of the connected enterprise

Results: lower inventory, reduced lead times, improved customer service, and better quality

By John Nesi

According to the Organization for Economic Co-operation and Development (OECD), the global middle class will surge from 1.8 billion in 2009 to 3.2 billion in 2020. This ballooning middle class will translate to greater consumer spending, placing an increased demand on manufacturing, resources, and infrastructure.

Meeting this demand requires automation investments to help operations be more productive, sustainable, and flexible. As global pressures for improved productivity and competitiveness continue to increase, companies need to invest in new, innovative ways to optimize plants and supply networks.

Fortunately, the next industrial revolution is happening right now, and the connected enterprise is the reason for this shift. It is converging information technologies (IT) and operational technologies (OT) and holds the promise for improving global production, sustainability efforts, and overall business agility. If companies are not on board with this industrial shift, they will fall behind. It is easy to understand the implications of this shift, but it is more challenging to justify the automation and cultural investment to leverage it.

A case in point

At automation manufacturer Rockwell Automation, about a third of the 22,000 employees work within 17 global manufacturing plants, which manage nearly 400,000 stock-keeping units and produce thousands of build-to-stock, configure-to-order, and engineer-to-order products. As an evolution of its decades-long commitment to capturing enterprise data to make better decisions, the company implemented an enterprise-wide strategy and automation investment several years ago to better connect its global manufacturing facilities and accelerate the business value of those connections.

The automation investment included a new approach to its manufacturing, focused on the following outcomes:

- A standardized, global information system and processes to enable accurate measurements across multiple site locations, including Asia, Latin America, Europe, and North America
- Contextualized, working data capital that employees and partners can use to make better decisions
- Enhanced collaboration of both systems and talent
- An information network optimized for the enterprise
- Security designed-in from the onset

Change motivators

Many manufacturers have a diverse product portfolio with plants spread across the globe executing a variety of manufacturing processes. That diversity creates added complexities and underscores the importance of standardization.

FAST FORWARD

- The next industrial revolution is happening right now, and the connected enterprise is the reason.
- Standardized, global information systems and processes enable accurate measurements across multiple site locations.
- Integrated automation investments have higher returns when early planning includes collaborative input from multiple stakeholders.

In the case of Rockwell Automation, each plant was running on its own enterprise resource planning (ERP) system and had its own custom applications that captured and analyzed data in different ways. It needed to implement a standardized approach across its global plants to gain consistent points of reference and processes for quality control, purchasing, and manufacturing engineering—regardless of location.

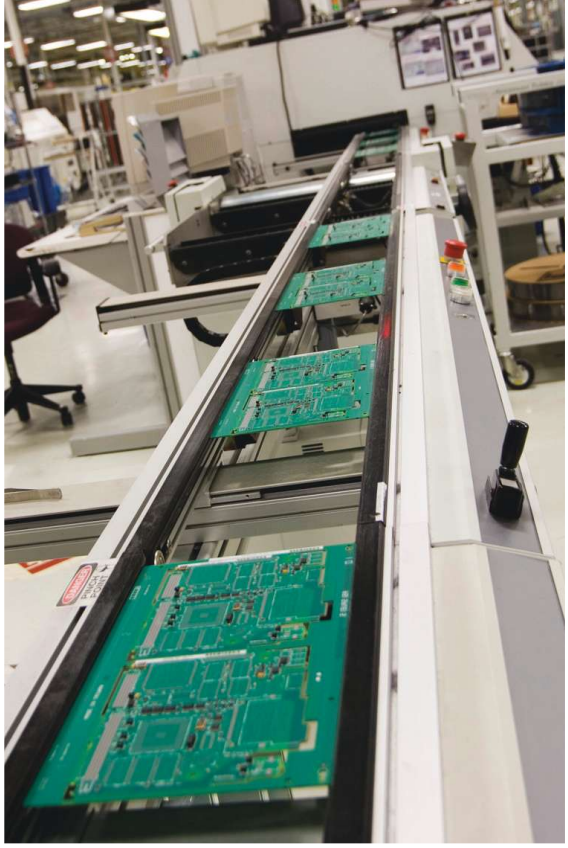
It also needed to decrease the hundreds of applications that were registered through its business process mapping. This large amount was due to the variety of products and parts manufactured by the company globally. Variation in data and input/output points collected throughout the system increased the chance of error and the amount of time needed to read and understand the collected information—a challenge not uncommon for large, global manufacturers.

In addition, having a connected system across the globe would allow the company to quickly respond to issues anywhere. For example, if there was a big surge in demand in China, its facility in Ohio would have visibility into its Singapore plant to respond and address production needs, and vice versa.

Planning for the connected enterprise

The company developed a five-year plan to entirely restructure its facilities and supplier network. To address and improve the plan's technology component, the company focused on updating the different manufacturing execution system (MES) solutions throughout its plants—each customized with little to no integration across the enterprise. By integrating applications into one system, overall production would improve and support the company's transition toward standardization.

Reducing variation and increasing information visibility at each site would enable managers and operators to measure production efficiencies and inefficiencies. It also would provide an all-encompassing view into the



company's operations and a company-wide benchmark to measure success.

Justifying the automation investment included understanding the shortcomings of the current MES solution at each site and how the company could make improvements to gain the best solution possible in tandem with its ERP system.

The company used an MES application that could be implemented throughout the enterprise. This solution was a cost-effective development platform with an extensible workflow engine and operational model that could grow with operations. The flexible solution also catered to the company's various product manufacturing styles.

The MES solution and enterprise manufacturing intelligence software now track and record data and pinpoint production trends. They fit the needs of individual sites but also have enterprise-wide capabilities. The solution acts as a funnel for multiple data sources, feeding out understandable, actionable working data capital to make improvements. It pulls information from hundreds of applications, streamlines it into one centralized location, and feeds it into the ERP system.

The scalable solution also analyzes key performance indicators (KPIs) in real time to measure quality, consistency, and process efficiency. More specifically, the solution has the following production-related capabilities:

- Production dispatching: Tracks work order management, confirms ERP orders, and provides back-up assistance for ERP system outages
- Production quality: Monitors out-of-bounds conditions in real time, managing rework and defects
- Production workflow management: Monitors verification points on each product during production, delivering compliance and regulatory reporting
- Production performance: Collects test and measure data, providing an enterprise benchmark

Justifying automation investments in greenfield plants

When the company began construction on two greenfield plants in Monterrey, Mexico, it underwent a business requirements analysis to identify all the layers of interactivity between plant equipment and the ERP system. With nearly 3,000 unique products manufactured at the two facilities combined, including printed circuit-board assemblies, motors, drives, power supplies, and light curtains, connection between each layer was vital for success.

Because the plants were new, the team had the opportunity to implement the MES solution simultaneously with the ERP system. The company configured and extended a comprehensive MES application that would integrate communication between the plant floor and the enterprise.

The new system has data collection capabilities that significantly improve product quality. For example, information for each step of the process must be gathered, managed, tracked, and made visible to plant operators so that engineers can identify areas of inefficiency, downtime, or diminished quality within the process. Rather than relying on each station on the line to create its own documentation, the new MES solution collects and sorts millions of data points in a systematic, more usable way. If a particular printed circuit-board assembly, for example, consistently fails a quality check, plant engineers can now use that data to improve the process or product design.

The new system creates a set of applications that can be used in all the company's major manufacturing facilities. It is a common manufacturing platform that can be expanded to different regions and different product groups, all while retaining an efficient and accurate way of measuring KPIs. It facilitates continuous improvements. For example, the Monterrey teams adjusted the MES solution to better align with its global workforce management software. With that integration, plant managers can more closely track labor against jobs completed; measure labor costs, time and attendance, scheduling, and absences; and show both successes and opportunities for growth.

The context-driven system also presents work instructions and operator information in English or Spanish, allowing new employees in the Monterrey plants to be trained within 30 minutes. Because there is only one system to learn, operators can be easily cross-trained in other functional areas of the plant.

The MES platform also excels at feeding data in and out of the ERP system, which reduces issue-resolution times consistently and supports leaner operations. That improves output efficiencies, the key to any manufacturer's profitability in building products.

Justifying automation migration investments in older plants

Justifying automation investments within older plants has unique challenges. Most manufacturers are trying to maximize their current asset investments while minimizing the production disruptions of any upgrades. For example, there is an older plant within the company's global manufacturing sites that produces a wide variety of complex products, approximately 2,500 different products each year. It needed better information

from the plant floor to enable operators to make more informed decisions to maximize efficiency.

The company replaced the existing, legacy MES that was becoming difficult to manage and maintain with a new, user-friendly, customizable solution. It allowed the team to add and remove equipment, revise layouts, and ensure that the solution fit the plant's needs. The real-time display of metrics and performance in relation to expected output gave managers a more efficient way to measure success. The system detects quality issues and provides feedback immediately, allowing managers and operators to address issues quickly and deliver feedback upstream.

Changing the older plant layout to accommodate the new system called for a change to its network topology approach, as well. The company used EtherNet/IP—a standard, open, single-network infrastructure that facilitates secure interoperability with corporate networks and industrial applications. It enabled a wealth of information beyond what plant operators had anticipated. This increased network traffic and required improved network segmentation, so the plant team started managing network structure and hierarchy by using the reference architecture framework. They can maintain real-time network performance and add capacity in the future with minimal impact to the network infrastructure.

Before the new network infrastructure, periodic delays due to data collisions would appear to operators as efficiency declines, but they could not see the cause. Now, with the new infrastructure and connected solutions in place, operators do not need to dig for the "why?" The tools provide validation and allow workers to be predictive rather than preventive and only estimating efficiency based on past experiences.

Operators and engineers can better understand the differences in products, realistic output, cycle times, manufacturing processes, and expectations. They also can focus on the equipment, products, and processes to improve productivity rather than manually gathering data from each line.

Assessing security risks while justifying automation

Risk—including security risk—is a critical consideration when justifying automation investments. With network convergence and the connection of previously disparate systems, security risks naturally increase. Because of this, security programs should be done collaboratively and holistically across all operations, rather than as a tacked-on solution.

For all the benefits of connecting the company's plant and manufacturing assets, engineering leaders understood how those connections introduced greater internal and external threats—both malicious and accidental. They very deliberately designed in security from the beginning, as opposed to implementing it after the system was installed.

The defense-in-depth security approach used is multilayered—both physical and electronic defenses—to stop threats at multiple levels using numerous safeguards within the manufacturing zone, including device, controller, process, and enterprise. Security program assessments included vendors and suppliers, which were evaluated in the same way as the company's own organization. A single technology or methodology simply will not suffice against the multitude of vulnerabilities that exist.

Return on the connected enterprise investment

Tangible returns on the automation investment include lowering inventory metrics 31 percent, capturing 30 percent each year in capital avoidance, improving supply chain delivery to 96 percent, reducing lead times by 50 percent, improving customer service metrics such as time-to-want to 98 percent, and reducing parts-per-million quality issues by 50 percent. The company estimates its productivity has improved four to five percent annually.

The internal delivery team and plant managers have already made plans for further optimizing the automation investment:

- Continuously improve connectivity throughout the supply chain to prevent defects and minimize scrap
- Leverage IT to improve efficiencies and drive productivity
- Gather, understand, and apply data so that operators can read and convert data into actionable information much more intuitively
- Flexibly expand the MES solution, so it can be applied to a wider variety of products and processes around the world

Automation investments that enable continuous connected enterprise improvements support manufacturers' needs to constantly improve production speed, efficiency, and quality; generate insights from real-time visibility and historical trending; better manage inventory; and improve tracking and tracing capabilities.

Holistic automation investments like this have even higher returns when early planning includes collaborative input from multiple stakeholders. Operations, IT, and engineering need to share what they are trying to establish, discuss implications for each role, and detail needed information and output goals. As the planning expands to include more roles throughout the enterprise—adopting new goals along the way—manufacturers will uncover an even stronger justification to invest. ■

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John Nesi (janesi@ra.rockwell.com) is the vice president of market development for the global sales and marketing segment of Rockwell Automation. Nesi has more than 33 years of experience in sales, marketing, systems, services, operations, and engineering across various businesses with the company. He is responsible for the strategic and commercial development of growth initiatives, including connected enterprise, sustainability, power and energy, working with customer segments in the industrial sector to promote energy efficiencies, process optimization, and production intelligence to improve demand-driven production in a safe, sustainable environment. Nesi holds a bachelor's degree in electrical engineering from Cleveland State University and is an executive scholar of Kellogg School of Management.

View the online version at www.isa.org/intech/20150206.

Steps to navigate the industrial Internet journey

By Bernie Anger

ABOUT THE AUTHOR

Bernie Anger is the general manager of GE's Intelligent Platforms Business, a business unit within GE Industrial Solutions. He is driving the transformation of the company's automation portfolio in support of the industrial Internet. A native of Argentina, Anger studied economics at Universidad de Buenos Aires and earned an MBA from Universidad Torcuato di Tella.

In today's connected world, machines can communicate and coordinate with each other in ways that were not possible before. These brilliant machines are changing the way businesses work, enabling them to optimize efficiencies, boost productivity, reduce costs, and bring products and services to market faster. This major transformation is powered by the industrial Internet—which combines advances in computing and data collection to deliver advanced analytics to industry.

Optimizing your current operations

By connecting your machines and industrial big data, the industrial Internet enables you to better understand your operations. You can visualize how your assets and processes are behaving in real time and know what events occurred. What's more, you can drive operations to their fullest potential based on key parameters such as market demand, weather conditions, plant environment, and load balancing.

Say you are a mining company and you need better visibility into your complex production processes to best control your operations. Solutions powered by the industrial Internet can, for example, help you monitor your processes and control loops. When a control loop is insufficient, you can quickly identify what changed and why. You can get to the root cause of the issue and gain better insight into your process and control problems to minimize deviations from set parameters.

The result: greater consistency, process optimization, and safe production.

In addition, to capture the full potential of optimization, you not only need to understand the current state of your operations, but also be able to predict what will happen. Although industrial machines have always issued early warnings, they were inconsistent and presented so that details often crowded out the most critical information. The advent of networked machines with embedded sensors and advanced analytics tools has changed that reality. Now you can leverage advanced analytics in context to predict, providing real-time operational intelligence to the people who need it to make the best decisions. Thus, you have intelligent foresight to take appropriate action and proactively avoid issues—a powerful capability that allows continuous operation.

Industrial Internet maturity model

To help companies take the industrial Internet journey, we developed the industrial Internet maturity model. It helps you assess where your organization falls on the spectrum of expected maturity or capabilities within the industrial Internet and discover the best solution pathway to achieve your business priorities.

There are five key steps in the model that build on each other to help you optimize performance:

1. Connect: Provides the foundation to connect your machines and lets you collect data from your assets and processes and manage that data to derive value.
2. Monitor: Focuses on helping you understand the performance of your assets and processes and visualize events.
3. Analyze: Helps determine the root cause of issues based on historical and real-time data so you understand relationships, correlations, and trends, and can effectively troubleshoot problems.
4. Predict: Focuses on providing foresight into impending problems so you can avoid issues and drive greater process consistency and asset uptime.
5. Optimize: Maximizes the performance potential of your assets and processes to achieve your desired outcomes and leverage the benefits of the industrial Internet.

The value of the model is that it is a solid solution path toward optimization. The assessment can be done across various levels—from your core assets and adjacent assets, to across your facility and enterprise. You can understand where your business is today in terms of adopting the industrial Internet and determine where you want to be in the future.

Ultimately, the industrial Internet allows you to combine brilliant machines with best-in-class analytics to deliver valuable new insights. The results of these insights can empower your business to avoid unplanned downtime, maximize profitability, and transform your technology infrastructure.

Forward-looking companies have begun to embrace the industrial Internet to drive new performance highs and generate value like never before. They are leveraging connectivity and analytics to achieve their business priorities—whether it is increasing throughput, improving product quality, driving resource efficiency, shortening response times, or other valuable outcomes.

The industrial Internet is a journey—one that can lead businesses to a new wave of productivity gains and performance optimization. And it has meaningful implications for us all. ■

Renewable energy research facility power monitor

By Jennifer Palumbo

National Renewable Energy Laboratory (NREL) scientists and engineers research renewable energy and energy-efficient technologies, often partnering with private industry to deliver these ideas to the general public. NREL designed the Energy Systems Integration Facility (ESIF) to facilitate research in solar and wind, grid planning and operations, energy storage, building technologies, fuel cells, and advanced vehicles.

Each lab at the state-of-the-art 182,500 square foot facility is designed and instrumented to foster research on energy systems integration. The labs' capabilities include performing hardware in-the-loop simulations at megawatt-scale power, petascale computing (performance in excess of 1 petaflop, i.e., 1 quadrillion floating point operations per second) at the facility's High Performance Computing Data Center, and the unique research electrical distribution bus. The bus functions as a power integration circuit that can connect multiple sources of energy (such as solar inverters) with laboratories and experiments. It consists of two alternating current (AC) and two direct current (DC) ring buses that interconnect multiple sources of energy across the facility, giving NREL plug-and-play testing at grid-scale levels.

Because the ESIF is intended for researching various forms of energy integration, NREL researchers needed a system for monitoring power usage and controlling safety systems for each experiment or laboratory on a facility-wide scale. Monitoring AC and DC power loads gives researchers power usage information for their individual experiments and allows facility managers to view the total energy usage for the whole facility or for each laboratory.

The challenge

Researchers needed a power monitor they could configure for each experiment for multiphase AC and DC measurements. They also needed a configurable power monitor they could use to mimic multiple types of circuit breakers while analyzing power characteristics such as real power, reactive power, and energy. For live analysis of complex energy experiments, researchers required high-speed and high-resolution data (16-bit, 50 kilosamples per second [kS/s]). Finally, the power monitor needed to communicate to various third-party human-machine interfaces and programmable logic controller (PLC) devices. A system integrator was engaged to design, engineer, and integrate a power monitor with these features and capabilities.

The solution

The team considered using traditional PLCs during preliminary phases of the project; however, it quickly became apparent that the system performance requirements could not be achieved with this technology. Key factors included high-speed sample rates and demanding real-time processing. These requirements were satisfied with an engineered solution based on a programmable automation controller. It combined embedded real-time computing and field-programmable gate array (FPGA) technology to meet performance requirements, along with high-performance input/output modules. Software images were developed for more than 70 programmable automation controllers needed for the application and installed within electrical panels throughout the ESIF.

Visual integrated development environment software was developed for the controller's CPU and the onboard high-performance FPGAs to deterministically perform required functions in parallel threads for speed. With data acquisition greater than 50 kS/s, the system integrator implemented applications to efficiently transfer high-speed data from the FPGA to the real-time application processor.

The solution monitors electrical conditioning components between the ESIF's power sources (120 V, 240 V, 480 V, 600 VAC, and 1,000 VDC on 250 A and 1,600 A buses) and the laboratory power connections. As a result, any powered device under test or equipment used for testing in the lab is actively monitored. Voltage transformers bring voltage levels into a range compatible with controllers and protect equipment from excess voltage.

Using a programmable automation controller that combined embedded computing with FPGA technology was critical to the project's success. It allowed the team to implement a system that identifies unsafe electrical conditions and reports them back to the control systems. Simultaneously, the controller's sample data across input channels using high-speed analog input modules combined with a global positioning system time-stamping-and-synchronization module that guarantees timing synchronization between controllers is accurate within 100 nanoseconds. The high performance of the controllers allows the systems to acquire and process 28 channels at 51,200 hertz while managing other functions, such as alarm monitoring, TCP/IP, and Modbus communications. This was especially beneficial when correlating data from multiple laboratories, where testing areas are separated and monitored by different devices within the large test facility.

The system integrator developed and compiled more than seven custom FPGA application functions for this project. The individual compile time for each FPGA application averaged 1.5 hours. Furthermore, before releasing final versions of the FGPA applications, several preliminary compilations were necessary to account for project scope changes and shifting requirements. The system integrator improved productivity using the control vendor's LabVIEW FPGA Compile Cloud Service to compile the FPGA applications. The cloud service allowed developers to push multiple FPGA builds to an external server compiling code in parallel. This freed up developer machines for other tasks, making compile times 2.6 to 5.3 times shorter. ■

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Grand Challenges for Engineering

Throughout history, engineering has advanced civilization from how people connect and heal to how we get around and have fun. But society still faces major challenges. The National Academy of Engineering has outlined 14 game-changing opportunities for the 21st century called the Grand Challenges of Engineering. These challenges, falling under the themes of health, sustainability, security, and joy of living, are:

- make solar energy economical
- provide energy from fusion
- develop carbon sequestration methods
- manage the nitrogen cycle
- provide access to clean water
- restore and improve urban infrastructure
- advance health informatics
- engineer better medicines
- reverse-engineer the brain
- prevent nuclear terror
- secure cyberspace
- enhance virtual reality
- advance personalized learning
- engineer the tools of scientific discovery

Engineers, students, and the general public are requested to review the challenges and then create and submit a one-to-two minute video showing how achieving one or more of the challenges will lead to a more sustainable, healthy, secure, or joyous world. The competition is open to individuals or teams in four competition categories.

The challenges are an opportunity for automation engineers to promote their profession. Patrick Gouhin, executive director and chief executive officer of ISA, urges automation professionals "to be more vocal about the role automation professionals can play in solving the world's 'Grand Challenges.' By design, these multidisciplinary challenges will, if solved, better humanity forever. And we play a critical role in solving these!"

The National Academy of Engineering is offering a grand prize of $25,000 to the most inspiring video. There is also a People's Choice Award of $5,000, and the top videos in each category are eligible for a prize of up to $5,000. Videos will be judged based on the creativity in the content selection and presentation, the anticipated breadth of public appeal, and the effectiveness in highlighting how achieving the challenges will accomplish one of the themes. The entry deadline for submission is 2 March 2015. Contact E4Uvideocontest@nae.edu or www.e4uvideocontest.org for more information. ■

ISA FPID Symposium

ISA's second annual Food and Pharmaceutical Division (FPID) Symposium took place at press time in Philadelphia. This was an outstanding opportunity to learn world-class technical content, interact with subject-matter experts, explore the latest products and services, and participate in valuable networking activities. Food and pharmaceutical process industry professionals learned how to improve control, optimization, measurement, and serialization and earned continuing education credits and professional development hours.

Government officials were invited to deliver updates and clarify regulatory requirements.

Expert-driven technical program and topics

A diverse array of technical presentations covered the very latest news, trends, and projections about products and process. Government officials were invited to deliver updates and clarify regulatory requirements. Topics the conference addressed include:

- Automation business value
- Electronic record systems
- Manufacturing execution systems and batch execution systems
- Automation, disposables, and the factory of the future
- Serialization case study with a global perspective
- Using data to optimize the product life cycle
- Overall equipment effectiveness
- Mitigating cybersecurity risks

ISA training courses

Attendees earned continuing education credits by attending one of the two optional ISA training courses being held in conjunction with the conference.

- Introduction to Applying the Batch Control Standard ANSI/ISA88.01 (IC40C)
- Implementing Enterprise/Control Integration Using ANSI/ISA95 Standards (IC55C)
- Training took place 9 February from 8 a.m. – 4 p.m. off site at Applied Control Engineering, Inc., 2075 Quaker Pointe Drive, Quakertown, Penn.

Renowned keynote presenters

Two visionary and highly honored innovators—Peter Martin, Ph.D., vice president of business value solutions at Schneider Electric, and Paul McKenzie, Ph.D., vice president of manufacturing and technical operations at Janssen Pharmaceutical Companies (a subsidiary of Johnson & Johnson)—delivered keynote presentations at the event.

For more information about the 2015 FPID Symposium, including the program and schedule, visit www.isa.org/FPID2015 or call ISA at +1 919-549-8411. ■

ISA Certified Automation Professional (CAP) program

Certified Automation Professionals (CAPs) are responsible for the direction, design, and deployment of systems and equipment for manufacturing and control systems.

CAP question

Common terms used to quantify dangerous failures include which of the following?

A. probability of failure on demand (PFD) and nuisance trip rates
B. probability of failure on demand (PFD), risk reduction factor (RRF), and safety availability (SA)
C. mean time between failure, spurious (MTBFsp); nuisance trip rates; and safety availability (SA)
D. mean time between failure, spurious (MTBFsp) and risk reduction factor (RRF)

CAP answer

The correct answer is *B*, probability of failure on demand (PFD), risk reduction factor (RRF), and safety availability (SA). Spurious trips and nuisance trips are indicative of "safe" failure modes, not "dangerous" failures. This makes answers A, C, and D incorrect.

PFD, RRF, and SA are extremely important in quantifying dangerous failures.

- Probability of failure on demand: A value that indicates the probability of a system failing to respond to demand when a failure occurs.
- Risk reduction factor: RRF = 1/PFD
- Safety availability: SA = 1 – PFD

Reference: ANSI/ISA-84.00.01-2004 standard

ISA Certified Control Systems Technician (CCST) program

Certified Control System Technicians (CCSTs) calibrate, document, troubleshoot, and repair/replace instrumentation for systems that measure and control level, temperature, pressure, flow, and other process variables.

CCST question

A circuit has a 100 Ω resistor, a 50 Ω resistor, and a 200 Ω resistor all in parallel with each other. **What is the total resistance of the circuit?**

A. 28.6 Ω
B. 35.0 Ω
C. 175 Ω
D. 0.28 M Ω

CCST answer

The correct answer is *A*, 28.6 Ω. To find the total resistance for resistors in parallel, use the equation:

$$\frac{1}{Rt} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3} + \cdots$$

For this problem:

$$\frac{1}{Rt} = \frac{1}{100} + \frac{1}{50} + \frac{1}{200}$$

$$\frac{1}{Rt} = 0.035$$

$$Rt = \frac{1}{0.035} = 28.6\ \Omega$$

Reference: Goettsche, L. D. (Editor), *Maintenance of Instruments and Systems, Second Edition*, ISA, 2005.

Basics of continuous level measurements

By Gene Henry

The most common technologies available for continuous level measurement are ultrasonic, free space radar, guided wave radar, capacitance, gamma, and pressure. Let's take a look at each technology and some typical applications.

Ultrasonic

An ultrasonic transducer generates a mechanical sound pulse that is directed through air to the process. When this pulse encounters the process surface, it bounces back to the transducer (figure 1). The transmitter is basically a high-tech timer, measuring the time it takes the pulse to travel to the process and back. This time is directly proportional to the distance to the process surface.

This time-based technology is referred to as time of flight (ToF). Ultrasonic transmitters are used on a variety of simple applications for measuring liquid or solid level in a vessel. The transducers are temperature compensated to give a high level of repeatable accuracy at distances of 2 to 230 feet.

Applications with heavy dust or high temperatures are not well suited for ultrasonic transmitters. If there is too much dust in the air, it defuses the signal and causes a poor return. High temperatures or vapors can also alter the density of the air enough to affect the speed of the pulse transmission, causing errors in measurement.

Mechanical signals travel at the speed of sound and require an atmosphere to transmit the signal; therefore, ultrasonic devices do not work in applications operating under a vacuum. Foam on top of a liquid can also disrupt an ultrasonic signal. The acoustic signal can be absorbed by the foam, resulting in no return echo.

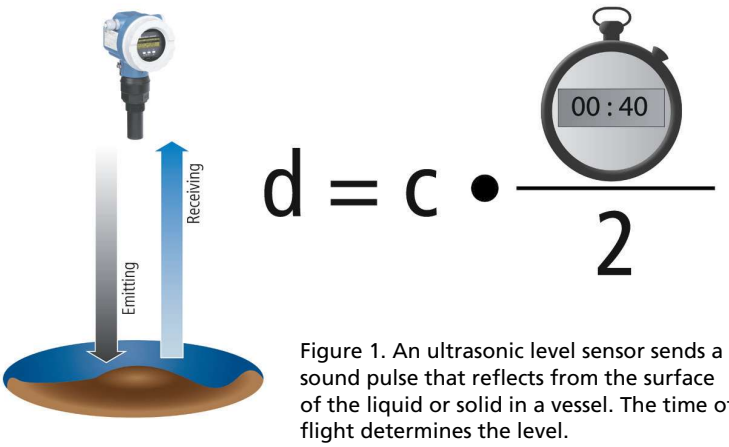

Figure 1. An ultrasonic level sensor sends a sound pulse that reflects from the surface of the liquid or solid in a vessel. The time of flight determines the level.

Free space radar

This noncontact radar technology has two different versions: pulse generated and frequency modulated continuous wave (FMCW). The pulse-generated version works on a ToF principle similar to an ultrasonic device. An electromagnetic wave between 1 and 100 GHz is sent from the antenna toward the process surface in search of a change in impedance, which will reflect the signal back to the transmitter. In most cases, the difference in the dielectric between air and the process material will cause the signal to be reflected back. The dielectric of the product is important when selecting a radar unit because the greater the dielectric, the greater the change in impedance and the stronger the reflection. The size of the radar horn (figure 2), the dielectric of the product, and the condition

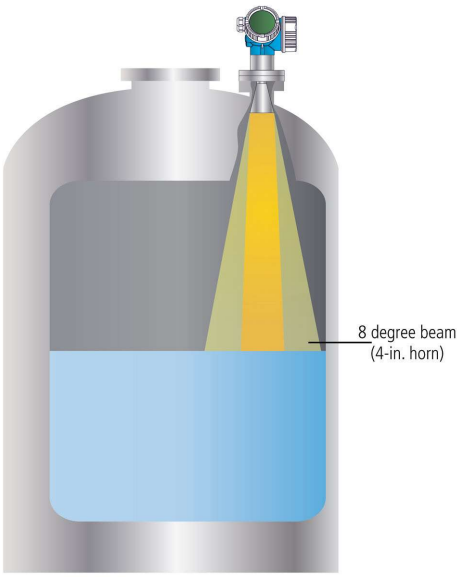

Figure 2. Free space radar sends an electronic signal that either bounces off the surface or shifts its frequency to determine level.

of the process surface (calm or agitated) determine the maximum distance from the device to the process surface.

The FMCW radar versions send out a continuous radar signal, and the frequency shifts as the distance to the process changes. Because FMCW is a continuous wave, it never loses touch with the material, making it better for agitated vessels.

Free space radar is relatively unaffected by environmental conditions such as different gases or vapors between the transmitting device and the process surface. It is largely impervious to variations in process temperature or pressure, and it can work in a full vacuum.

Two process conditions that can affect free space radar are condensation on the transmitter antenna and foam on a liquid surface. Condensation is typically a high dielectric liquid, and the radar signal cannot penetrate this material, resulting in increased "noise" in the launch area of the signal. Foam is very difficult to quantify for radar as it is not readily distinguishable from the process surface.

Even with these limitations, free space radar is the most universal noncontact level technology, and it will work in most liquid or solids level applications.

Guided wave radar

Guided wave radar or time domain reflectometry (TDR) works very similarly to pulse-generated free space radar. The main difference is the addition of a cable or rod from the radar unit to the process surface to guide and focus the radar signal (figure 3). Guided radar also operates on a lower frequency of approximately 1.2 GHz.

The advantage of guided wave radar technology is the signal is very concentrated on the cable or rod. When it encounters foam, the radar signal does a better job of going through the foam to reflect off the liquid surface. Neither condensation nor dust has any effect on TDR.

Another major advantage in solids level measurement is that the angle of repose can be accounted for with careful placement of the rod or cable. The angle of repose results from the way solids pile up in a vessel, creating an angle on the side of the pile. When using guided wave radar, the point at which the rod or cable contacts the product will determine the signal reflecting back to the transmitter.

Installation considerations include material compatibility, possible excessive pull force on the cable in solids applications when installed in tall silos, and avoidance of interferences between the cable or rod and items such as agitators.

Capacitance

A capacitance level transmitter works much like a capacitor. The metal center rod of the probe acts as one plate of the capacitor; the vessel wall acts as the second plate. If the material being measured is nonconductive, it acts as the dielectric insulator between the two plates. If the material being measured is conductive, the insulation on the probe is the dielectric insulator in the capacitor. The vessel is essentially turned into a variable capacitor.

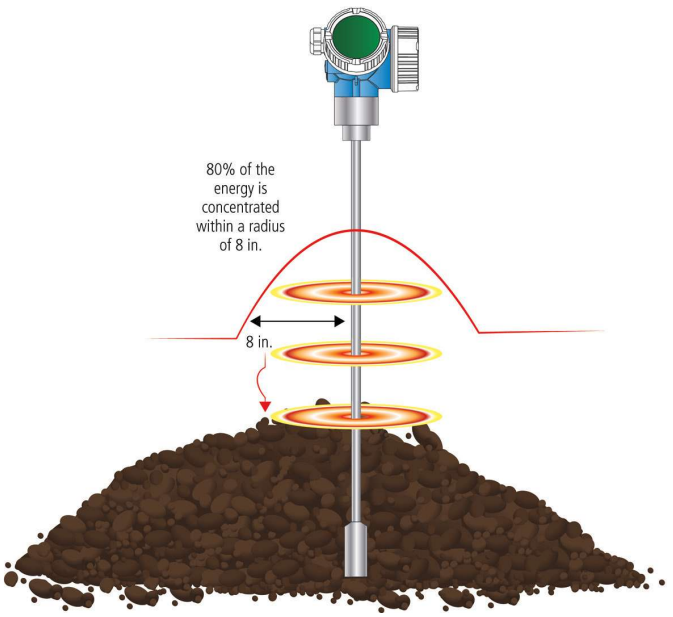

Figure 3. The signal from a guided wave radar transmitter follows a cable or rod to the process surface for improved performance.

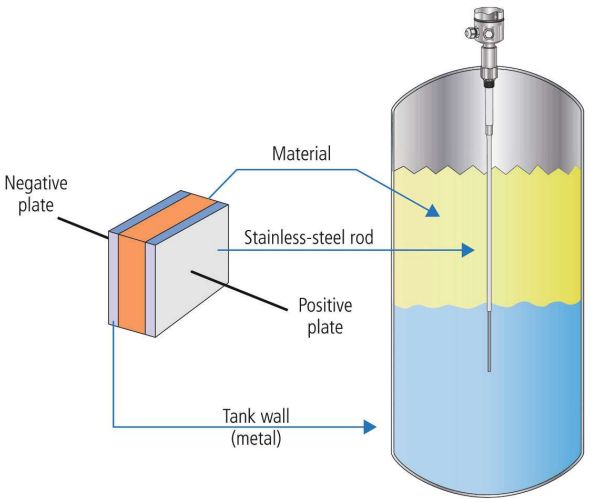

Figure 4. Capacitive level sensors measure many kinds of products.

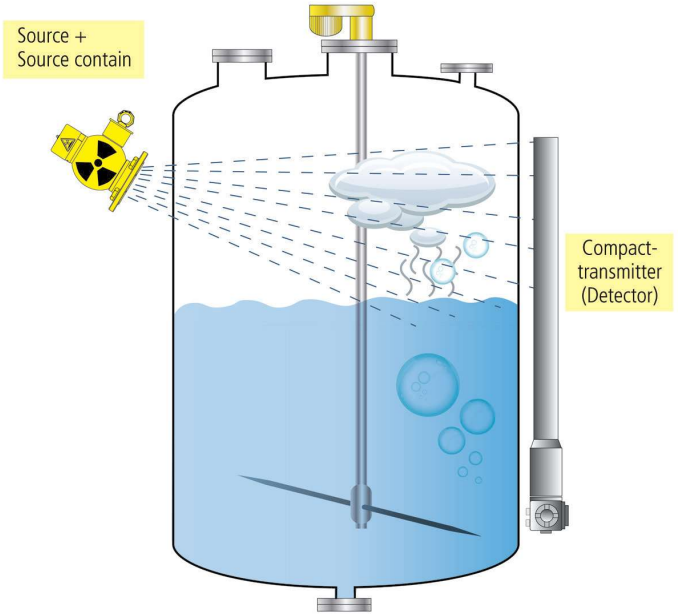

Figure 5. Gamma sensors can measure level in harsh environments.

The more product in contact with the probe, the higher the capacitance expressed in a picofarad (pF) value reading. This pF value is a measure of capacitance and relates to the height of the product.

The advantages of a capacitance level transmitter are the speed of response and the wide variety of products that will work with this technology. It is used extensively in filler bowl applications, oil reservoirs, and short chemical storage vessels. Capacitance is also the most reliable form of electronic level measurement for interface applications where there is a combination of conductive and nonconductive liquids forming layers within the vessel. When using capacitance level measurement, it is necessary to have a metallic vessel or a ground reference rod for the transmitter to work correctly.

Gamma

This technology consists of two components: a radioactive source and a detector (figure 5). The radioactive source is about the size of a dime and is mounted in a lead-filled source container that has only a small lens or opening to direct the gamma radiation in one direction. The gamma energy is created by the decay of a radioactive source, which is normally cesium-137 or cobalt-60. This gamma energy is absorbed or partially blocked by anything between it and the detector.

In a continuous level application, when the vessel is full

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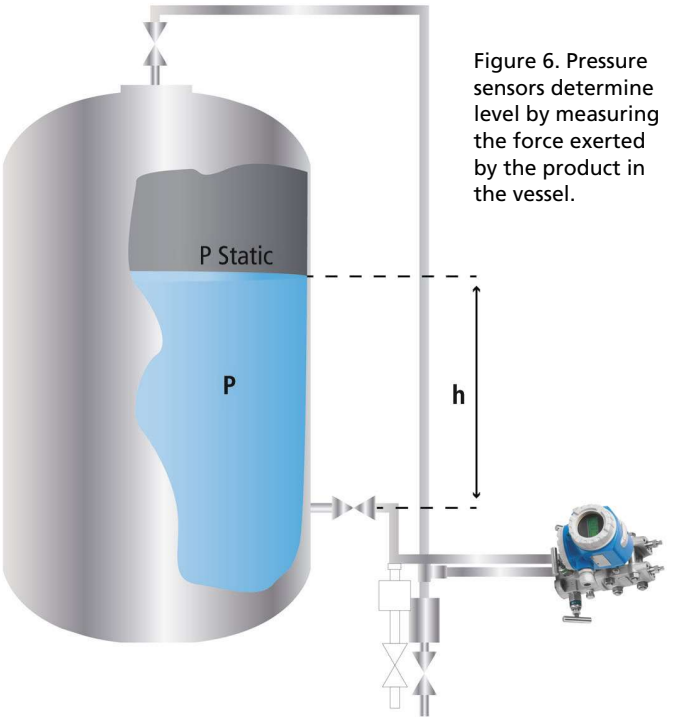

Figure 6. Pressure sensors determine level by measuring the force exerted by the product in the vessel.

only a very small amount of radiation is detectable on the opposite side of the vessel where the detector is mounted. The amount of radiation increases as the vessel level drops. Gamma can be used for point level, continuous, and interface measurements. This technology is also used for density measurements where other technologies such as Coriolis flowmeters are not suitable.

This technology can be used for the harshest environments including high temperature, corrosive media, high pressures, and toxic or abrasive materials. Gamma systems are used on reactors, separators, and mixers of all types. The main disadvantages of the technology are the regulations and licensing requirements that an end user must follow with the radioactive material.

Pressure

Pressure has been used to measure level for many decades. Older systems used a differential pressure transmitter with a constant pressure regulator and a bubble tube. Any increase or decrease in the height of the liquid in the vessel was reflected by an increase or decrease in the back pressure measured in the bubble tube.

Newer style hydrostatic pressure transmitters make this same direct pressure reading in nonpressurized vessels by having the pressure sensor mounted in contact with the process fluid at the bottom of the vessel, or by using a sensor mounted on a rod or cable, which is lowered into the vessel. This technology is commonly used in sumps, storage, and process vessels.

In a pressurized vessel (figure 6), a differential pressure transmitter must be used as the transmitter. It must subtract the head pressure from the pressure measured at the bottom of the vessel. The square root of the resultant measured value is proportional to the level of the associated column of process fluid. Any changes in the specific gravity or weight of the process fluid will affect accuracy.

No matter what technology you choose to measure level, always look at the accuracy you want to achieve and the process conditions. What is the pressure and temperature of your process? Will there be foam or gas fumes associated with this process? Is there an agitator or turbulent surface in the process? What is the dielectric or specific gravity of your process? In most applications, you will find there are multiple technologies that will work, with the final choice best determined by working with an experienced supplier of all the leading level measurement technologies. ■

ABOUT THE AUTHOR

Gene Henry (Gene.Henry@us.endress.com), senior level business manager for Endress+Hauser in the U.S., has more than 35 years of experience in the instrumentation field. He started his career as an instrument foreman in the phosphate mining industry, and has spent the past 20 years in instrumentation sales, providing consulting services to industrial and municipal plants. In his current role, he works closely with the sales force on level applications and in the development of marketing strategies for Endress+Hauser level products in the U.S.

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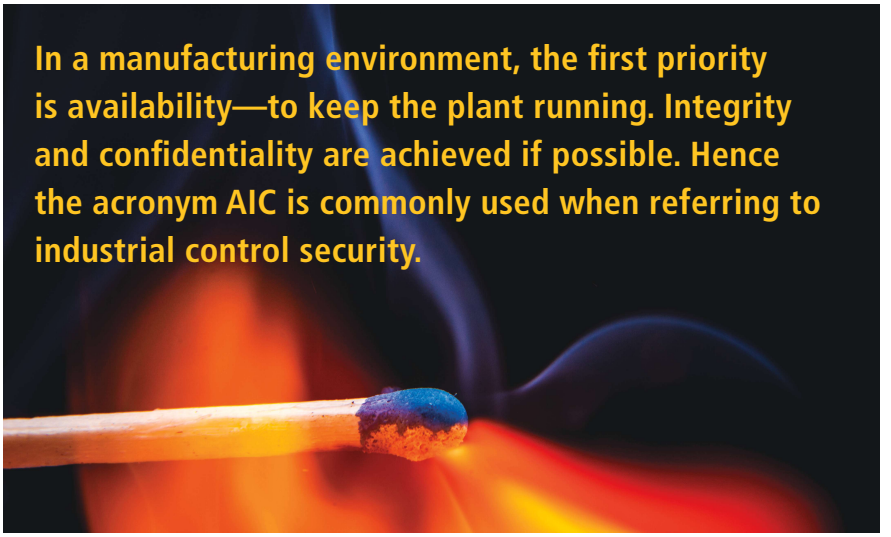
Bah humbug!

By Peggie Ward Koon, Ph.D.

One of my favorite shows is *NCIS Los Angeles*. Every Monday night I sit in front of the tube to see what type of threat agents Callen, Deeks, Kensi, Sam, Nell, and Eric will encounter. Usually these episodes are all about crimes being investigated, either in the U.S. or at some remote place around the globe, by a fictional team of the Naval Criminal Investigative Service (NCIS). And these agents intervene to either mitigate or eliminate the threats. One of the most interesting episodes aired during the holidays and was entitled "Humbug." The story began with the team investigating a case that at first glance appeared to be about a fire that was deliberately started in the server room of a cybersecurity company, See Bug Systems. See Bug Systems developed software to protect companies from cyberattacks. The company also produced malware that it used to test its software. And the owner of the company was being asked by the U.S. government to help prevent cyberattacks against U.S. mission-critical systems.

As the plot unfolds, the NCIS agents realize that the arsonists used the fire in the computer room as a diversion; the real threat is a file that was copied from the servers to a thumb drive. Eric explains that the thumb drive contains malware that can be used to shut down *programmable logic controllers* used in automation. This particular malware could be used to shut down electric power plants. And Sam explains that if the file got into the wrong hands, it would be disastrous. The file would allow its owners to shut down communications and electric power, resulting in no heat, no lights, and no electrical power—across the entire country.

When asked how this happened at a company that sells cybersecurity software, the owner said that a new security system was scheduled to be installed after Christmas. Of course there were many other twists and turns in the story, including the use of other malware to divert funds to a secret corporate account. There was deception, fraud, murder, and all the typical components of a classic episode of *NCIS LA*.



I really related to this episode for several reasons. First, it was all about *cybersecurity*—one of the hottest topics today. You can hardly watch, listen to, or read the news without encountering a story about cyberthreats. From the famous Sony Pictures hackers threat to identity theft at department stores and banks—cybersecurity is being discussed everywhere.

Second, it was about *automation*. More often than not, the cybersecurity discussion is focused on preventing threats related to identity theft, confidentiality, finances, intellectual property, or other types of enterprise data, information, or assets. But if you attend an automation seminar, participate in an automation webinar, or attend any conference on manufacturing, automation, or operational information technology (IT), a cybersecurity discussion or presentation is sure to be included as one of the tracks on the schedule. Cybersecurity is needed in automation, too!

Now I know that there is a plethora of television shows and movies focused on cybersecurity and cyberthreats. But this NCIS story is not just a story about cybersecurity, deception, fraud, and personal security breaches. In the *NCIS Los Angeles* episode, a fictitious company—like so many real companies—is not *secure*, because it failed to either identify and/or address both infrastructure and technology vulnerabilities. *The episode raises awareness of a different type of cybersecurity—the type that addresses a threat to our nation's mission-critical infrastructure—a threat that is real.*

Finally, the episode highlights the need for automation companies that provide critical services—IT, power, food and pharmaceuticals, water and utilities, manufacturing, chemical, oil and gas, and others—*to not only have secure systems and components, but also to have a workforce that has been trained to detect and deter cyberattacks* affecting mission-critical operations.

Cybersecure operations: Understanding the threat and developing the workforce is key

According to the ISA website (www.isa.org), "Cybersecurity for the industrial enterprise is quite different from cybersecurity for other areas." Enterprise security (whether for an office or even for a

bank or credit card processor), is usually focused on confidentiality, integrity, and availability. The acronym CIA is often used to emphasize that securing these functions of the enterprise is paramount and that protecting the data on the servers is the first priority.

Operations security, or industrial control security, is different. In a manufacturing environment, the first priority is availability—to keep the plant running. Integrity and confidentiality are achieved if possible. Hence the acronym AIC is commonly used when referring to industrial control security. "ISA has developed the most comprehensive standard, ISA99, which has now become the global industrial cybersecurity standard IEC 62443. Also, the ISA Web site has textbooks, training, and a new certification program. The ISA99/IEC 62443 Cybersecurity Fundamentals Specialist Certificate is the first of several certification programs designed to provide clear evidence that you have understood and been trained to work in the very different world of industrial control security and systems." Read more at www.isa.org/technical-topics/cybersecurity.

ISA is also working to secure control systems using ISASecure™ components and systems from the ISA Security Compliance Institute. "The ISA Security Compliance Institute manages the ISASecure™ program, which recognizes and promotes cybersecure products and practices for industrial automation suppliers and operational sites. The ISASecure™ designation is earned by industrial control suppliers for products that demonstrate adherence to an industry consensus cybersecurity specification for security characteristics and supplier development practices." Learn more about ISASecure™ and how to become part of the movement to secure critical infrastructure around the globe at www.isasecure.org.

Our government is also doing its part to raise cybersecurity awareness and to enhance our nation's preparedness both with regard to critical infrastructure and workforce development. For example, the Cybersecurity Enhancement Act of 2014 was passed by Congress and signed by President Obama in December. The bill provides for "an ongoing, voluntary public-private partnership to improve cybersecurity, and to strengthen cybersecurity research and development, workforce development and education, and public awareness and preparedness, and for other purposes." The Cybersecurity Enhancement Act of 2014 also formalizes the role of the National Institute for Standards and Technology in continuing to develop the voluntary Cybersecurity Framework. It includes provisions to promote cybersecurity research; private-public sector collaboration on cybersecurity, education, and awareness; and technical standards, which include a federal cloud computing strategy. It also directs the National Science Foundation to continue the Federal Cyber Scholarship-for-Service, a program designed to "increase and strengthen the cadre of federal information assurance professionals that protect the government's critical information infrastructure." Scholarship recipients "agree to work in the cybersecurity mission of a federal, state, local, or tribal agency for a period equal to the length of their scholarship." Read more at www.csee.umbc.edu/2014/12/cybersecurity-enhancement-act-of-2014.

Why is this important?

As the 2015 chair of the Automation Federation, I have become acutely aware of the importance of workforce development for mission-critical operations and the importance of providing cybersecurity training as a part of that effort. On a personal note, two new nuclear reactors are being built near my hometown. These reactors will be a critical economical source of energy for the people like me in the surrounding areas. And because these reactors will be computer controlled, they also must be secure from cyberthreats. Recently I had an opportunity to tour the facility, and one of the first questions I asked was related to cybersecurity, including workforce development, at the plant. It is important to me, my family, and my community that the reactors are secure from cyberattacks and that the workforce is trained to identify, assess, and take actions to prevent cyberattacks.

Likewise I try to remain aware of what our government is doing to help ensure that our nation's mission-critical infrastructure is secure from cyberattacks and that the workforce that supports those systems is properly trained, so that they understand and have the tools to detect and deter cyberattacks against those mission-critical operations.

What should you do?

Don't say "Bah humbug!" If you are an automation professional or an employee, manager, or owner at an automation company, you will want to make sure that you and your workforce understand cybersecurity for mission-critical operations and that you and your teams know how to ensure that your operations are secure. If you are affiliated with an educational institution that trains personnel in mission-critical operations, you will want to understand what courses and resources are required to provide cybersecurity training for mission-critical operations. Find out more about cybersecurity and developing secure operations and a cybersecurity-trained workforce at www.isa.org/technical-topics/cybersecurity. ■

ABOUT THE AUTHOR

Peggie Ward Koon, Ph.D., (pkoon@peggiewardkoon.com) is the 2015 chair of the Automation Federation, the 2014 president of ISA, and the former vice president of audience at the *Augusta Chronicle* and Chronicle Media, a division of Morris Publishing Group, Morris Communications.

ISA-95 standard to reduce time to integrate business and operational systems

The ISA95 committee has completed the sixth standard in the widely used Enterprise-Control System Integration series with the publication in late 2014 of ANSI/ISA-95.00.06-2014, *Enterprise-Control System Integration Part 6 – Messaging Service Model.*

"The newly released Part 6 provides a standard set of services for the exchange of messages between systems," stated Dennis Brandl, ISA-95 series editor. "Integration of business systems and operational systems involves the exchange of messages as defined in ISA-95 Parts 2 and 4."

"Normally these messages are exchanged using an enterprise service bus or message exchange middleware," Brandl continued. "Every vendor has developed its own interface for sending and receiving messages. Part 6, the message service model, defines a standard interface that systems can use to publish messages, subscribe to publications, send a query message, and listen for response messages. Implementations of Part 6 will significantly reduce the time required to integrate business and operational systems. The services are defined so that they can be layered onto existing enterprise service buses and message exchange middleware, providing a standard for application development. Organizations and companies are working with the ISA95 committee to provide a standard set of Web service definitions for the Part 6 services."

On to Part 7

Completion of Part 6 followed the publication of two major ISA-95 revisions in 2013:

- ANSI/ISA-95.00.03-2013, *Enterprise-Control System Integration Part 3 – Activity Models of Manufacturing Operations Management*
- ANSI/ISA-95.00.05-2013, *Enterprise-Control System Integration Part 5 – Business-to-Manufacturing Transactions*

The committee is now focused on developing a Part 7 standard that will be based on the use of ISA-95 object models defined in Parts 2, 4, and 5 (Parts 1 and 3 do not contain object models) to define a set of services and messages that may be used to exchange namespace element definitions between communicating applications. Applications may have, and often do have, different identifications for the same objects. In any exchange, all parties to the exchange must agree on the names or identifiers of the objects. Often this agreement is managed through a set of equivalent identifier registration services. There are many different kinds of registration services, each usually focused on a particular requirement and for a particular namespace.

Implementations of Part 6 will significantly reduce the time required to integrate business and operational systems. The services are defined so that they can be layered onto existing enterprise service buses and message exchange middleware, providing a standard for application development.

Part 7 will define an equivalent identifier service model and formal object model for translating identifiers in Level 3 to Level 4, and Level 4 to Level 3 exchanges. As such, Part 7 will define a simple implementation approach to a subset of the much broader issue that is addressed by ISO/IEC 11179 Information Technology – Metadata registries (MDR) and ISO/IEC 19763 Information Technology – Metamodel framework for interoperability (MFI).

Integrating ISA-95 and ISA-88 workflows

Beyond the Part 7 work, the ISA95 committee is in the early stages, in conjunction with ISA88, Batch Control, of addressing the integration of ISA-88 and ISA-95 workflows. When the ISA-88 and ISA-95 standards are applied together in an application or project, some of the terminologies, models, and key definitions in these two sets of specifications need to be aligned to assist users. The envisioned ISA88-95 technical report is intended to enable an integration team to use these specifications jointly, in an application or project. The purposes would be to:

- Explain how ISA-88 and ISA-95 workflows interrelate
- Explain to practitioners how to use ISA-88 and ISA-95 workflows together
- Improve understanding about the ISA-88 and ISA-95 series by addressing such questions as when they are both needed, when one is more applicable than the other, and which one should be used

Committee review comments on the Part 7 draft and the ISA-88/95 workflows will be covered at an ISA95 meeting 11–13 March 2015 in Dallas, Texas, U.S. The meeting will immediately follow and partly overlap a meeting of IEC/ISO JWG 5, which adopts the ISA-95 series as the IEC/ISO 62264 series. JWG 5 is a joint effort of IEC SC65E, Devices and Integration in Enterprise Systems, and ISO TC 184 SC 5, Interoperability, Integration, and Architectures for Enterprise Systems and Automation Applications. Dennis Brandl is the JWG5 convener, and several other ISA95 leaders actively participate in the joint work.

JWG 5 reflects a long and productive collaboration by which many original ISA standards have been adopted as IEC global standards in vital areas including industrial cybersecurity (ISA-99), batch process control (ISA-88), functional safety (ISA-84), management of alarm systems (ISA-18), and wireless systems for automation (ISA-100).

For information about viewing or obtaining the ISA-95 series of standards, visit www.isa.org/standards. For information about participating in the ISA95 committee, contact Charley Robinson, crobinson@isa.org. ■

Focus on level

dP level transmitter

The Deltabar FMD71 level transmitter uses two pressure sensor modules, each connected electronically to a single transmitter. Using a Ceraphire ceramic sensor in the pressure sensor modules, the transmitter calculates the differential pressure from both sensors and transmits the level, volume, or mass via 4–20 mA with HART as a standard two-wire loop-powered device.

One sensor module measures the hydrostatic pressure and the other one the head pressure. Each sensor sends a digital signal corresponding to temperature-compensated measured pressure to the transmitter. This electronic system eliminates impulse lines and capillaries and their related issues of icing, clogging, leaks, condensation, and changing ambient temperatures.

The sensors connect to the transmitter module via industry-standard, color-coded twisted-pair cable and have NEMA 4X/6P (IP66/IP68) watertight housings and connections. The transmitter can be located away from the sensors in areas safe and convenient for personnel. The system meets ATEX, IEC Ex, CSA, and FM specifications for use in hazardous areas including Zones 0, 1, and 2 and Divisions 1 and 2.

Endress+Hauser, www.endress.com

Submersible level transmitters

The ProSense SLT series submersible level transmitters are for applications such as lift station monitoring, construction bypass pumping, tank liquid level, wastewater, and slurry tank liquid level control. Available in two styles, the submersible level sensors measure the liquid level by continually sensing hydrostatic pressure produced by the height of liquid above the sensor. The 4–20 mA output signal is compatible with programmable logic controllers, panel meters, data loggers, and other electronic equipment.

The SLT1 series has a 1-inch diameter housing and a ported bullet nose cap to protect the sensor diaphragm. Starting at $299.00, this series is available in cable lengths ranging from 30 feet to 140 feet and with sensing ranges of 0–5 psig to 0–50 psig. The SLT2 series has a 2.75-inch diameter PTFE flexible diaphragm surrounded by a 316 stainless-steel nonfouling protective cage for reliability in harsh environments. Starting at $470.00, this series is available in cable lengths ranging from 30 feet to 100 feet and with sensing ranges of 0–5 psig to 0–30 psig.

Both styles are fitted with a shielded cable containing an atmospheric vent tube and a tough polyurethane jacket that incorporates a "water block" liner beneath the jacket. The cable is attached to the sensor using an over-molding process to prevent moisture intrusion.

Automation Direct, www.automationdirect.com

Guided wave radar level transmitter

The SmartLine guided wave radar level transmitter can measure the level and volume of liquids inside processing tanks up to 160 feet high. The transmitter has an application validation tool that allows customers to input technical data about their specific process tank. They can also validate that the correct level transmitter application is delivered to the site ready to install out of the box.

The level transmitter has a choice of a basic alphanumeric display or an advanced graphics display. The advanced graphics display, capable of showing process data in graphical formats and communicating messages from the control room, also offers multiple languages and full integration with the company's Experion process control system.

Honeywell, www.honeywell.com

Miniwave level transmitter

The Miniwave is a loop-powered ultrasonic level transmitter that continuously measures liquids up to 39 ft (12 m) deep. The 2-inch transducer offers high chemical compatibility, which makes it suitable for multiple applications in all industries. Installation and configuration are simplified with flexible mounting and the push button on the front face. There is a four-line graphic display for level indication and configuration. The device has a powder-coated cast aluminum housing and a glass window to view the display. The unit comes standard with HART, 4–20 mA output.

Hawk, www.hawkmeasure.com

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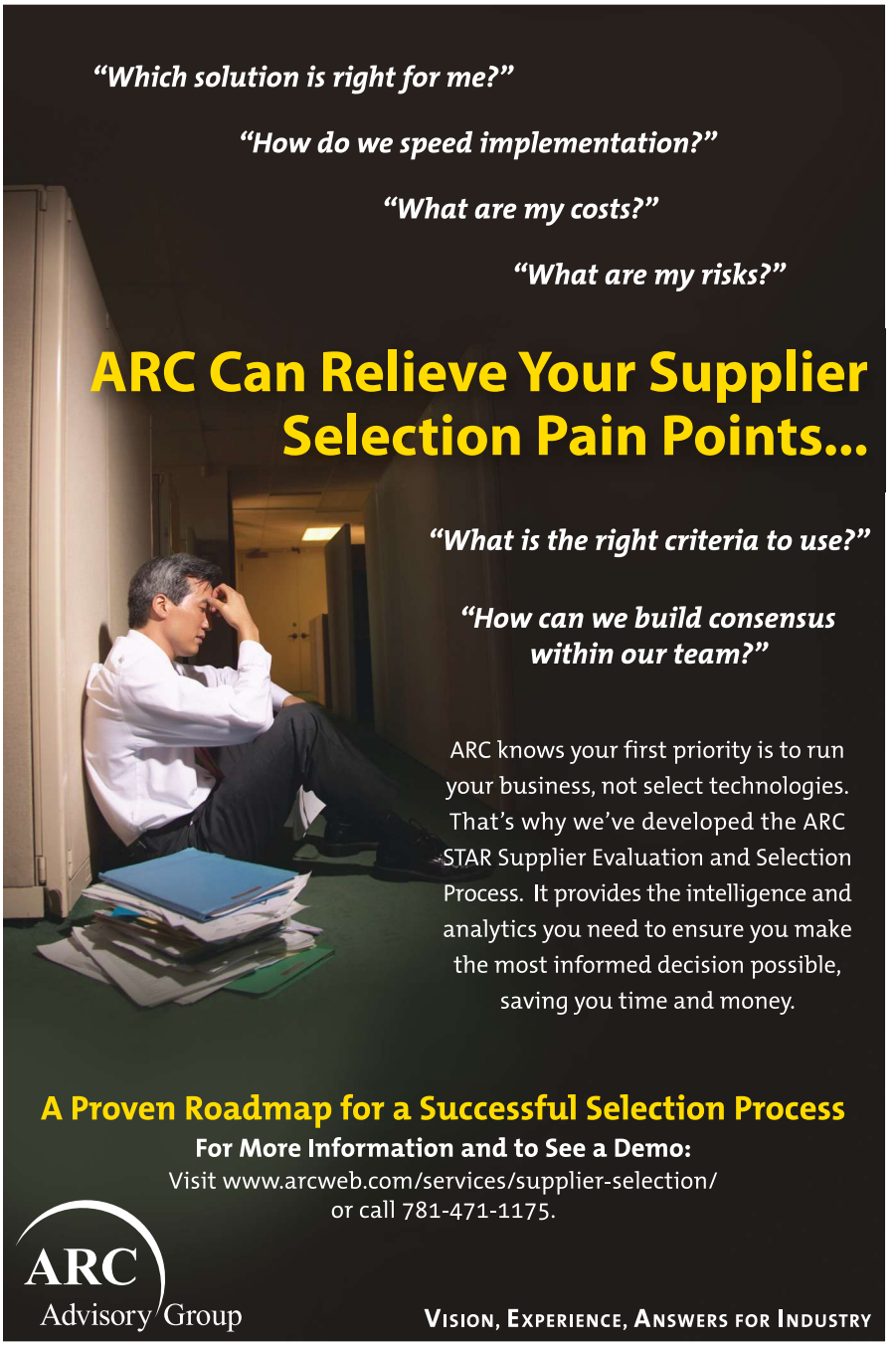

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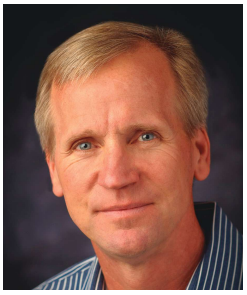
Advanced control as operations support

By Allan Kern, P.E.



ABOUT THE AUTHOR

Allan Kern, P.E., (kern@linandassociates.com or allan.kern@APCPerformance.com), is an advanced controls consultant with Lin & Associates, Phoenix, Ariz., and is the founder of APCPerformance.com, an online advanced control resource. Kern is a professional control systems and chemical engineer with 35 years of industry experience. He has authored numerous papers on practical process control solutions, on topics ranging from field instruments to expert systems.

In operating plants, process engineers, also known as production or operation engineers, are usually heavily engaged in day-to-day operations support. They are more likely to be found in their hard hats, in ad hoc meetings, or working with operators than sitting at computer terminals. Their job description typically involves a good understanding of process economics, process constraints, and taking care of the equipment in their charge.

I am often surprised at how little process engineers and control engineers know about each other's jobs, especially as they share the same underlying goal of process optimization. Today's control engineers are often more occupied with the health of the control system than the state of the process. And process engineers often do not realize that a modern control system is a flexible and powerful tool—more like a robotic transformer than a pump—that can be brought to bear on a wide range of operating issues.

As control systems have modernized, control engineers have moved away from operation support and into a wide range of control system support, information technology, and reporting activities. Now industry needs to backfill the control engineering role that is more focused on applying the control system than supporting it. Leaving this role unfilled handicaps the operating team and neglects the automation potential of a modern distributed control system.

Process control and automation constitutes a core competency in the process industries. To meet modern manufacturing needs in safety, reliability, and agility, an operating plant must maintain a level of in-house control and automation capability that can serve as a ready resource and regular contributor to the operating team.

That is easier said than done. The loss of traditional control engineering know-how due to downsizing and retirement has long been lamented. But I think the problem is due to management decisions and industry trends as much as structural factors. Huge resources have gone into multivariable control, which unfortunately has failed to deliver the kind of automation success and operating agility that the process industries should be shooting for.

Multivariable control has probably moved industry farther away from effective operation support. Outsourcing means that resources have flowed out of the company, while more practical advanced control know-how has dramatically diminished from neglect and resource starvation. Multivariable control performance routinely undershoots expectations, and is likely to continue that way, based on its own emerging structural limitations. And the large controllers now deployed in industry often act as obstacles to agility, rather than as solutions, even as they continue to take a bite out of available support resources.

The recent era has left industry with control engineers who are quite skilled in control system support, but weak in operation support. Industry can address this by shifting a number of strategies:

- Control engineers should know process economics, process constraints, and the opportunities to improve them, just like process engineers. They should work as a team, not be strangers.
- Skilled control engineers should focus on supporting the operating team, not the control system or multivariable control.
- Many control system support activities lend themselves well to outsourcing. Outsource these activities, not advanced control.
- Traditional advanced regulatory control (ARC), which includes basic techniques, such as feedforward and adaptive gain, as well as more sophisticated solutions, such as custom programs and sequential controls, is where the best automation opportunities often reside. ARC took a back seat to multivariable control long ago, but it may be time to put ARC back in the driver's seat.
- Advanced control resources should be used to build in-house ARC competency and a "can-do" automation culture of agility, targeted solutions, and incremental successes, rather than a monolithic strategy.

Process control and automation in general (not multivariable control in particular) is the key technology of our era when it comes to process improvement of many kinds. It is a core competency of the process industries and warrants involved management and effective resource allocation. Industry should commit to building a higher level of in-house competency and manufacturing agility as a strategy to take process control and automation to a higher level of success in industry. ■

RESOURCES

"Lose the pyramid"
www.hydrocarbonprocessing.com/Article/2856603/Lose-the-pyramid-and-find-process-control-success.html

"Multivariable control performance"
www.isa.org/intech/20140802

Small-matrix model-less multivariable control
www.chemengonline.com/tag/advanced-process-control

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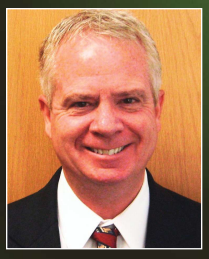

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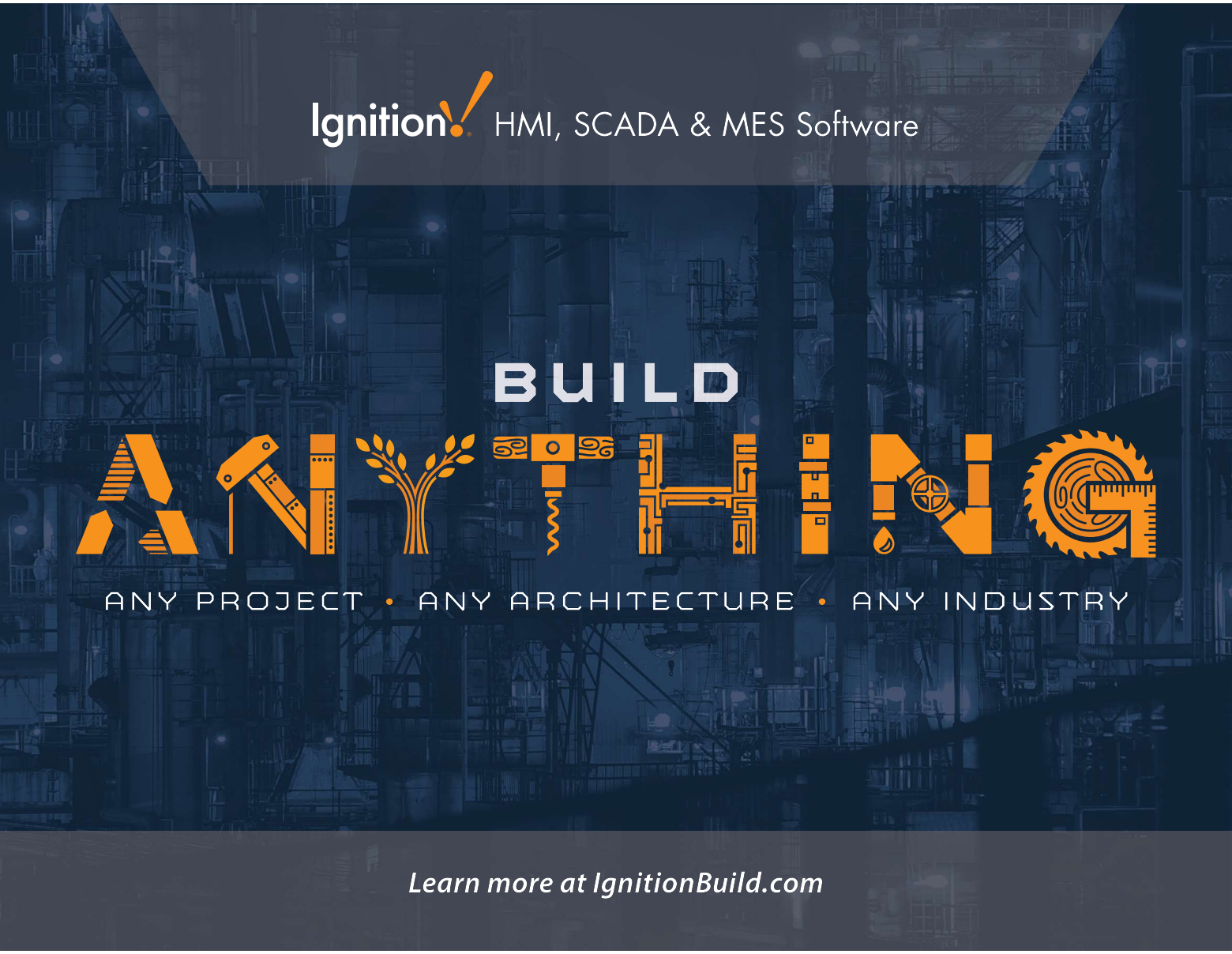
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